Excess Bond Premiums Around the Globe: A Cross-Country Database and Analysis[☆]

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Abstract

This paper introduces and analyzes a novel database of Excess Bond Premiums (EBP) for 41 countries over the period 1998–2023. Consistent with the literature on the "Global Financial Cycle", we document a strong common component in country-level EBPs and show that global financial conditions are driven primarily by U.S. monetary policy, with domestic monetary policy not even playing a significant role in shaping their own domestic financial conditions. Our findings support lender-based theories of global risk premia, whereby dollar funding costs are transmitted through wholesale funding markets to the global banking system, affecting credit provision worldwide. Despite the strong common component in financial conditions, country-specific EBPs retain significant predictive power for future regional economic performance, providing additional information for economic forecasting and inference.

Keywords: Excess Bond Premium, Financial conditions, Global Financial Cycle, Monetary policy, Global spillovers

JEL Codes: E44, E52, F34, F42

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1 Introduction

This paper presents, discusses, and analyzes a new database of Excess Bond Premiums (EBPs) for 41 countries between January 1998 and June 2023, applying the approach of Gilchrist and Zakrajšek (2012) to a multi-country setting.

As a measure of financial conditions, the credit spread has been proven to be highly valuable. For the U.S., Gilchrist and Zakrajšek (2012) document that the credit spread, and in particular the component of the spread not related to default risk – the so-called "EBP", thought to capture the risk-bearing capacity of financial intermediaries – is a strong indicator of financial conditions and is highly predictive of future economic development. Since then, the U.S. EBP has been central to forecasting models worldwide, also proving to be a useful control variable when trying to analyze the effect of monetary policy shocks through Vector Autoregressions (VARs) or Local Projections (LPs). While financial conditions worldwide are clearly strongly interlinked, it remains an open question whether the U.S. EBP sufficiently captures financing conditions outside the U.S., or whether a domestic or truly global indicator could provide further valuable information.

While the correlation between country EBPs is high, we document meaningful cross-sectional variation. Both the EBP's global factor and its domestic components provide valuable information for forecasting real economic activity. The predictive value of country-level EBPs for future economic dynamics suggests that these variables should be included when using empirical methods (such as VARs or LPs) to conduct inference on economic policy.[1] We furthermore show that across countries, a tightening in country-specific financial conditions (as captured by that country's EBP rising) is associated with a reduction in output, inflation, equity prices, and long- and short-term interest rates, as Gilchrist and Zakrajšek (2012) found for the U.S. We also find that domestic financing conditions on the corporate bond market are primarily driven by U.S. monetary policy. Strikingly, in the euro area and the U.K., the two other economies for which we have high-frequency monetary policy surprises, domestic surprises do not affect even the domestic EBP.

The strong global transmission of U.S. monetary policy has been studied empirically and theoretically. Miranda-Agrippino and Rey (2020) highlight the strong response of financial constraints worldwide to U.S. monetary policy decisions. Globally, asset prices decline,

[1] Gertler and Karadi (2015) made this point in their analysis of the effects of U.S. monetary policy shocks, noting how the U.S. EBP is likely to summarize information of variables otherwise left out of the LP or VAR system.

banks deleverage, and risk aversion appears to increase in response to a U.S. monetary tightening, leading to rising credit spreads. Interconnectedness in the international banking sector has been found to play a crucial role in determining cross-border liquidity spillovers. Bruno and Shin (2015b) formalize the idea of cross-border transmission of financial conditions through bank leverage in an international banking system. Foreign liquidity shocks propagate through internal capital markets of banks with global operations while active liquidity management insulates banks from domestic shocks (Cetorelli and Goldberg (2012)). The degree to which local and foreign shocks affect financial conditions worldwide is still uncertain and has important economic implications.

This paper contributes to the literature on the global transmission of U.S. monetary policy. Using global granular bond-level data, we document that U.S. monetary policy surprises transmit to the EBP worldwide, shaping financing conditions abroad. The U.S. surprises affect financial conditions abroad primarily through the component capturing the *price of risk* rather than through direct *quantity of risk* changes in the non-financial sector, supporting previous literature on the international banking sector. This repricing is common to bonds of every currency and issuer nationality, and it is larger for riskier borrowers while leaving their default risk unchanged. On top of it, a within-firm analysis identifies a smaller surcharge on the dollar bonds of non-U.S. issuers that is absent from the dollar bonds of U.S. issuers, suggesting that the dollar matters not through the currency of the bond but through the financial intermediary holding it. While the importance of U.S. monetary policy for global financial conditions has been documented before, this study provides detailed evidence on the determinants of local bond market conditions and directly compares the importance of domestic and U.S. monetary policy. Our results have implications for the effectiveness of traditional monetary policy outside the U.S., as corporate financing is increasingly shifting towards credit markets.

To the best of our knowledge, this paper is the first to construct EBPs on a harmonized basis for a broad set of advanced and emerging economies, offering a new database that spans the major bond markets and currencies. This dataset might be of independent interest to researchers and policymakers. We find that our new country-level EBP contains valuable information on top of global and alternative domestic variables in forecasting exercises. We extend the findings of Gilchrist and Zakrajšek (2012) to a multi-country setting and document the importance of global and domestic financing conditions for forecasting economic activity.

The remainder of this paper is structured as follows. Section 2 explains how we construct our country- and global EBP series and summarizes the evidence supporting their credibility,

with a detailed comparison against the U.S. series of Gilchrist and Zakrajšek (2012) provided in the Appendix. Section 3 turns to the anatomy of the series: it documents a strong common factor in country EBPs – showing that corporate credit conditions are largely global rather than national – while establishing that the local component nonetheless carries economic content, as it helps forecast domestic macroeconomic outcomes. Section 4 studies the role of monetary policy, showing that global credit conditions respond primarily to U.S. monetary policy rather than to the policy of other major central banks. Section 5 exploits heterogeneity across firms, and across bonds of the same firm, to trace how U.S. monetary policy is transmitted to corporate credit conditions. Section 6 concludes.

2 Methods & Data Sources

This section sets out the construction of our EBP series. We proceed in three steps. We first describe the estimation methodology, which extends the framework of Gilchrist and Zakrajšek (2012) to a multi-country setting. We then detail the two data sources underlying the exercise – corporate bond spreads and firm-level probabilities of default – together with the sample construction and cleaning steps. We conclude by reporting the estimated credit-spread pricing regression that delivers the EBP.

2.1 Methodology

Our construction of the (Global) Excess Bond Premium – (G)EBP – series closely follows the approach of Gilchrist and Zakrajšek (2012) (GZ) for the U.S. In essence, the approach regresses the natural log of corporate bond spreads on the firm's probability of default (PD), bond characteristics, and a set of fixed effects. Following GZ we control for the age of the bond, its face value, remaining duration, and coupon rate, as well as a set of fixed effects (e.g., industry, rating, currency).

The only difference with GZ is that we are in a multi-country setup. In our baseline, we pool all bonds across countries and estimate a single set of coefficients that are common across countries, adding a country fixed effect to control for cross-country differences in average spreads.[2] That is, for each bond i issued by company j, in country k, we estimate

[2] As a robustness, we compute a country-by-country version in which each coefficient in the baseline regression is interacted with a country dummy, allowing all slopes to differ across countries.

the following at the monthly frequency:

$$\ln(S_{jik,t}) = \beta \log(PD_{jkt}) + \Gamma X_{jik,t} + \gamma_{ctry} + \gamma_{ind} + \gamma_{cur} + \gamma_{rat} + \epsilon_{jik,t} \tag{1}$$

where $S_{jik,t}$ is the Option Adjusted Spread (OAS) of bond i issued by firm j in country k at time t;[3] PD_{jkt} is the 12-month probability of default of firm j in country k at time t, and $X_{jik,t}$ is a vector of bond characteristics including the log age of the bond, its log face value, log effective duration, and log coupon rate. The fixed effects γ_{ctry}, γ_{ind}, γ_{cur}, and γ_{rat} capture country, industry, currency, and rating fixed effects, respectively. The error term $\epsilon_{jik,t}$ captures the idiosyncratic component of the spread not explained by the model.

The fitted component of the spread $\hat{S}$ captures the default risk-implied spread. The excess bond premium (EBP) is defined as the difference between the observed spread and its fitted component, widely interpreted to capture corporate bond market sentiment (or time-varying risk premiums, independent of firm characteristics). This is thought to reflect the risk-bearing capacity, or risk appetite, of financial intermediaries.

Using $\hat{\sigma}^2$ to denote the estimated variance of the error term $\epsilon_{jik,t}$, and $\hat{y}_{jik,t}$ the fitted value of $\ln S_{jik,t}$ from equation (1), the EBP for bond i issued by company j in country k is thus obtained as follows:

$$\hat{S}_{jik,t} = \exp\left(\hat{y}_{jik,t} + \frac{\hat{\sigma}^2}{2}\right) \tag{2}$$

$$EBP_{jik,t} = S_{jik,t} - \hat{S}_{jik,t} \tag{3}$$

We obtain the global EBP (GEBP) as the simple average across all outstanding bond-level EBPs in a given month.[4] Using n_t to denote the number of bonds outstanding at time t, we thus have that:

$$GEBP_t = \frac{1}{n_t} \sum_i EBP_{jik,t} \tag{4}$$

We obtain a very similar measure if we run a country-by-country version of equation (1), rather than pooling across countries. In this case, we allow the coefficients β_k and Γ_k to differ across countries. See Appendix C.

Finally, we can also obtain country-specific EBPs calculated as the simple average across

[3]The OAS is calculated as the yield difference between the corporate bond and a synthetic risk-free government bond with an identical payment schedule. This cleans the yield from the price effect of embedded call options, ensuring the comparability of the spread across bonds.

[4]This is analogous to GZ, who calculate the U.S. EBP by taking a simple average of all U.S. bonds in their sample.

all outstanding bond-specific EBPs in a given month and in a given country k:

$$EBP_{k,t} = \frac{1}{n_{k,t}} \sum_{i} EBP_{jik,t} \tag{5}$$

where $n_{k,t}$ is the number of bonds outstanding in country k at month t.

2.2 Data sources & Cleaning

To construct our multi-country EBP series, we rely on two data sources: the global corporate bond data from ICE Bank of America Merrill Lynch and the corporate probability of default (PD) from the NUS Credit Research Initiative (NUS-CRI).

Corporate bond spreads. We collect monthly data for the Option Adjusted Spread (OAS) of corporate bonds traded on the Bank of America Global Corporate Index (G0BC) and the Global High Yield Index (HW00) between January 1997 (when the series starts) and October 2025. Due to data availability, the monthly spread data consists of end-of-month observations between 1997 and 2012 and monthly averages thereafter. In total, we collect OAS data for around 86,000 bonds issued by around 10,000 firms. We enrich the bond information with details collected from LSEG and Capital IQ firm identifiers to allow a matching with firm-level variables including corporate PDs.

Default risk. As a measure of firm riskiness, we rely on the 12-month PD constructed and provided by the NUS-CRI. Usage of this measure has two major advantages. First, it is available on a cross-country basis (spanning 47 countries). Second, this measure is arguably more comprehensive than sole reliance on Merton's "Distance to Default"-metric (DD; Merton (1974)), which sits at the heart of Gilchrist and Zakrajšek (2012). However, DDs do not fully explain the cross-firm variation in PDs (see, e.g., Bharath and Shumway (2008), Campbell et al. (2008), and Duffie et al. (2007)), calling for the inclusion of additional covariates. This is exactly what the NUS-CRI database does (consistently across 47 countries). In particular, it provides forward-looking PD at different horizons using firm-level variables (including Merton's DD, but also variables reflecting firm profitability and liquidity), as well as aggregate macrofinancial ones (such as the short-term interest rate and a stock return index); see Duan et al. (2012) and CRI (2022) for further details.[5] Ultimately, we collect PD data for a sample of more than 75,000 listed firms, spanning 47 countries, between February

[5]Consistent with Gilchrist and Zakrajšek (2012), we focus on the default probability at the 12-month horizon.

1990 and June 2023.[6]

Data cleaning, sample choices, and summary stats. We exclude bonds that experience a change in the ownership over their lifetime (e.g., due to a merger or acquisition)[7] as our static firm link prevents a time-consistent matching with a corporate PD. Bonds in our sample are issued in seven major currencies: the Australian dollar (AUD), the Canadian dollar (CAD), the Swiss franc (CHF), the euro (EUR) and its predecessors, the British pound (GBP), the Japanese yen (JPY), and the US dollar (USD). To capture the full richness of the data, we do not filter bonds by currency of denomination. As is usual in the literature, we restrict the sample to senior bonds, issued by nonfinancial corporations with a face value above USD 1 million, a remaining maturity between 1 and 30 years, and a spread between 5 and 3500 basis points. We further require that a country contribute at least 8 distinct issuers and an uninterrupted spread-and-default-probability spell of at least 36 months, so that its bonds identify a pricing relationship rather than a handful of isolated quotations. Finally, we winsorize the option-adjusted spread and the log default probability at the 1st and 99th percentiles, taking the percentiles within each month and pooling across countries, so that the cap is time-varying and binds on issuers that are distressed relative to the rest of the world at that date rather than relative to their own country's history. Appendix B.1 states the filters in the order applied and discusses the winsorization choice.

On the default risk side, we observe multiple PD observations for multi-listed firms. We account for this feature by taking the average PD across listings to incorporate all firm information and to maximize coverage. We merge the bond-level data with the issuing firm's PD at the highest level of aggregation (the ultimate parent, or group, level), as the risk of default on a payment is likely to be primarily determined by the group's riskiness. The Capital IQ firm identifier link is used to combine the datasets. After merging and applying the filters above, the sample on which the pricing regression is estimated consists of 31,537 unique bonds issued by 3,931 firms, located in 41 countries. Appendix Table B.1 reports its coverage country by country.

Figures 1 and 2 plot the cross-sectional distribution of bond spreads and of firm default probabilities over time, both computed on this final merged sample.

Since our PD measure is only available for listed firms, and since the EBP-concept starts

[6]At the time of writing, later data are not available yet, as NUS-CRI only provides the data to academic researchers with a 2-year lag.

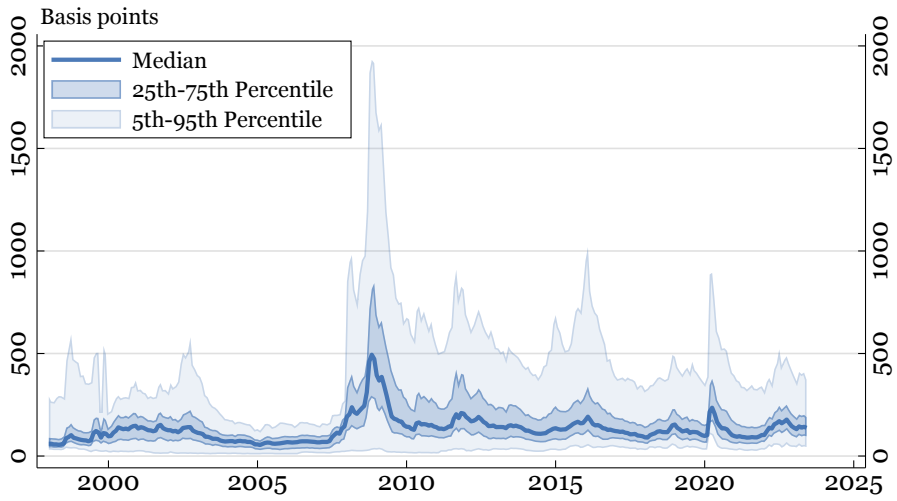
[7]We identify an ownership change when a bond's issuer name and ICE ticker change simultaneously; such bonds account for around 9% of observations.

Table 1 Summary Statistics for Bond Spreads and Default Probabilities

Variable	Mean	Std. Dev.	Min	Median	Max
Effective Yield (%)	3.96	2.83	-3.04	3.73	79.00
Spread (bp)	181.08	198.47	5.00	125.55	2986.00
PD-12m (%)	0.30	1.06	0.00	0.05	42.75
Coupon (%)	4.69	2.26	0.00	4.75	15.50
Amount Issued ($M)	633.80	536.85	18.45	500.00	15000.00
Maturity at Issue (Years)	12.65	8.69	1.12	10.00	50.00
Time to Maturity (Years)	8.64	7.61	1.00	6.01	30.00
Effective Duration	6.12	4.14	-5.55	5.00	26.38
Credit Rating (Composite)	14.78	3.03	1.00	15.00	22.00
Callable (% of Obs.)	37.08	48.30	0.00	0.00	100.00

Note. Summary statistics of the corporate bond spreads and default probability data used for the calculation of the EBP. Columns report the mean, standard deviation, minimum, median, and maximum across the bond-level monthly panel. The spread and the default probability are the winsorized series that enter the pricing regression (see Appendix B.1); the spread is in basis points, the default probability in percent, the amount issued in USD millions, and maturities in years. Sources: LSEG/BAML bond data and NUS-CRI default probabilities.

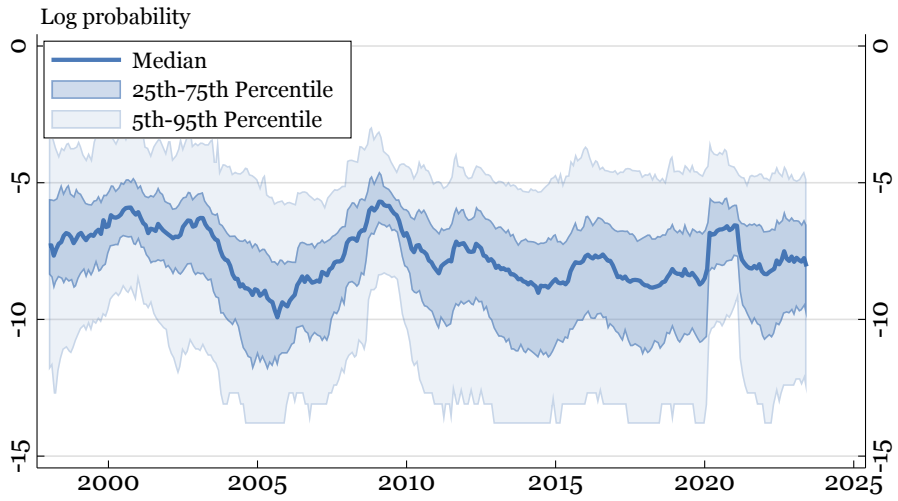
Figure 1 Cross-Sectional Distribution of Corporate Bond Spreads



Note. Cross-sectional median of the Option Adjusted Spread (OAS) of corporate bonds in the final sample, along with the 25–75 and 5–95 percentile ranges. The series plotted is the winsorized spread that enters the pricing regression; the winsorization acts only outside the plotted percentile range. The OAS is calculated as the yield difference between the corporate bond and a synthetic risk-free government bond with an identical payment schedule. This cleans the yield from the price effect of embedded call options, ensuring the comparability of the spread across bonds.

from firms having issued bonds, our final sample solely contains listed firms having accessed bond markets. Around 60% of the bonds in our sample are issued by firms located in the US, with these firms accounting for 54% of all firms in the sample. This reflects the fact that the

Figure 2 Cross-Sectional Distribution of Firm Default Probabilities



Note. Cross-sectional median of the log 12-month probability of default (PD) of firms in the final sample, along with the 25–75 and 5–95 percentile ranges. The series plotted is the winsorized log PD that enters the pricing regression; the winsorization acts only outside the plotted percentile range.

corporate bond market is dominated by U.S. issuances. At the same time, a substantial part remains distributed globally. The largest non-U.S. bond-issuing countries are Japan, the U.K., Canada, Germany, and France. By constructing an EBP for each of the 41 countries in our sample, we are able to provide a clearer picture of financing conditions worldwide.

There is also heterogeneity along the currency dimension. In the final sample, around 69.5% of bonds are denominated in USD, 21.0% in EUR, 3.6% in JPY, and 3.4% in GBP. The remainder is denominated in AUD, CAD, and CHF. Firms located in European countries predominantly issue in EUR, whereas USD issuance is most prevalent in most other countries. Issuance in Japan is highly tilted towards JPY (77.7%). The share of local currency issuance in the UK (22.8%), Australia (19.4%), and Canada (49.0%) is lower. Corporate bond issuance in emerging market economies is nearly completely denominated in USD.

Following all cleaning and merging steps, our final sample of bonds has a total notional outstanding of USD 7.2 trillion in 2023. The size of the total corporate bond market outstanding by non-financial corporations in 2023 is estimated to be around USD 22 trillion.[8] This shows that, despite the various cleaning steps, we manage to match a substantial part of the global corporate bond market to firm-level PDs. In our sample, firms have, on average, 4.7 bonds outstanding (see Table A.1 for country breakdown) with an average maturity at issuance of 13 years. The bonds cover the full spectrum of ratings from D to AAA. The

[8] LSEG

issuing firms' 12-month default probabilities are correspondingly dispersed, ranging from essentially zero to 43%, with a median of 0.05% and a mean of 0.30%.

2.3 Results

Table 2 summarizes the credit-spread pricing regression. Column (1) reports the pooled estimates, which constrain the loadings to be common across countries. Column (2) reports the mean-group estimator across the 41 country-specific regressions, which weights each country equally; the full set of country estimates is reported in Appendix Table C.1. Column (3) reports the U.S. regression on its own. Corporate spreads load positively on default risk and effective duration: both coefficients are significant at the 1% level in all three columns, and they are individually significant at the 5% level in 95% and 100% of the country-specific regressions respectively. The coupon carries a positive loading that is significant in 63% of countries, whereas bond age and face value are economically small and significant in only 27% and 41% of countries respectively.

The three columns agree on the sign of every loading, so the pricing of these characteristics is broadly stable across economies, but their magnitudes are sensitive to how countries are weighted. The pooled loading on default risk, 0.07, is roughly half the mean-group estimate of 0.13 and close to the U.S. estimate of 0.06; the same pattern holds for effective duration (0.31 pooled, 0.26 mean group, 0.32 for the U.S.). This is what one would expect given that U.S. bonds account for 53% of the estimation sample, so that the pooled specification largely reflects U.S. pricing while the mean group does not. The coupon is the exception: its U.S. loading of 0.36 exceeds both the pooled estimate of 0.16 and the mean-group estimate of 0.23.

Table 2 Credit-Spread Pricing Regression

	(1) Pooled	(2) Mean group	(3) US
Log PD	0.06***	0.12***	0.06***
	(0.01)	(0.01)	(0.00)
Log age	0.01***	0.02***	0.01*
	(0.00)	(0.01)	(0.00)
Log face value (USD)	-0.04***	-0.05*	-0.04***
	(0.01)	(0.02)	(0.01)
Log eff. duration	0.32***	0.28***	0.34***
	(0.01)	(0.01)	(0.01)
Log coupon	0.16***	0.19***	0.35***
	(0.03)	(0.03)	(0.03)
Observations	1,469,433	1,469,405	825,388
R^2	0.76	0.66	0.72
Currency FE	Yes	Yes	Yes
Rating FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Country FE	Yes	–	–
Regressions	1	41	1

Note. The dependent variable is the log option-adjusted spread; regressors are the log 12-month NUS default probability (NUS PD), log age, log USD face value, log effective duration, and log coupon, with currency, rating, and industry (ind4) fixed effects, and country fixed effects in the pooled regression only. Column (1) "Pooled" reports the single regression estimated on the full sample, with the loadings constrained to be common across countries; column (3) "US" reports the U.S. country-specific regression. In both, standard errors are two-way clustered by firm and month. Column (2) "Mean group" reports the mean-group estimator of Pesaran and Smith (1995) over the country-specific regressions, whose number is given in the *Regressions* row: the point estimate is the cross-country average of the country slopes, $\bar{\beta} = N^{-1} \sum_i \hat{\beta}_i$, and its variance is estimated from their cross-country dispersion, $\widehat{\mathrm{Var}}(\bar{\beta}) = \sum_i (\hat{\beta}_i - \bar{\beta})^2 / [N(N-1)]$, with normal-based inference. Standard errors are in parentheses; *, **, and *** denote significance at the 10%, 5%, and 1% levels. The *Observations* row reports, in column (2), the total across the country-specific regressions, which partition the same underlying sample as column (1); the two totals differ marginally because the finer country-specific fixed-effect partition leaves more singleton cells, which are dropped from the estimation. The R^2 in column (2) is the cross-country average of the per-country R^2.

3 An Anatomy of EBPs Across the Globe

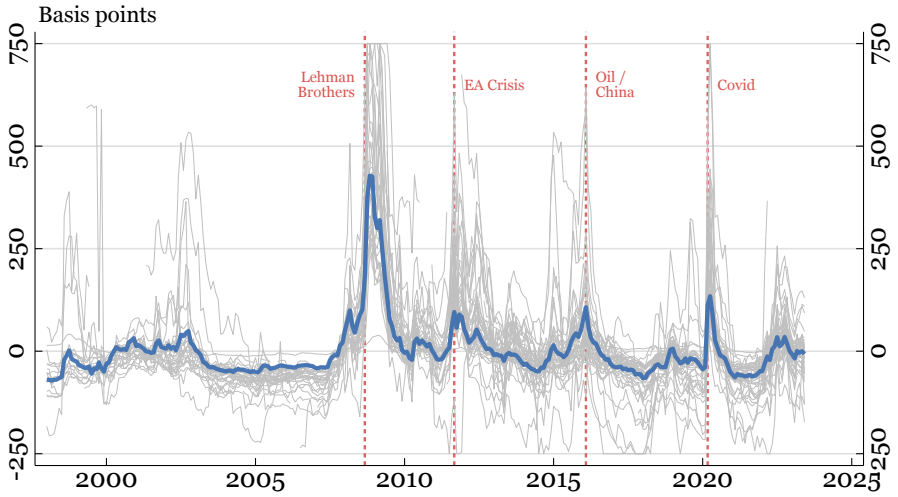
Having constructed the global and country-specific EBPs, we now examine their anatomy. The analysis proceeds in three steps. Section 3.1 describes the global series and the country series that underlie it, and shows that the global EBP tracks the major episodes of financial stress of the past quarter-century while individual countries differ considerably from one another. Section 3.2 shows that, notwithstanding that heterogeneity, a single common factor clearly dominates the panel – though it does not account for every country equally well.

Section 3.3 shows that the component the common factor leaves unexplained is not noise: it carries economic content, and it does so even for countries whose EBP the common factor tracks closely.

3.1 The Global EBP and its Country Components

Figure 3 plots the global EBP, together with the cross-country distribution of the country-specific (or local) EBPs.

Figure 3 Global EBP Country-specific EBPs



Note. global EBP (solid) against the individual country EBP series (thin grey lines). Monthly, January 1998 to June 2023, in basis points. Vertical dashed lines mark the collapse of Lehman Brothers (September 2008), the escalation of the euro-area sovereign debt crisis (September 2011), the trough in crude oil prices and the associated stress in high-yield markets (February 2016), and the onset of the COVID-19 pandemic (March 2020). All country series are plotted in full: no observation is clipped to the axis range.

The series tracks the major episodes of financial stress of the past quarter-century, marked by the vertical lines in Figure 3. Not surprisingly, the GEBP has its largest spike during the Global Financial Crisis, peaking at 428 basis points in November 2008, in the immediate aftermath of the collapse of Lehman Brothers. The low levels of the GEBP prior to the crisis show that it was preceded by a period of high risk bearing capacity by financial intermediaries (which may, in fact, have contributed to its causes; López-Salido et al. (2017)). Further peaks occur in September 2011 (96 basis points), at the height of the euro-area sovereign debt crisis; in February 2016 (107 basis points), when the collapse in crude prices and concerns over Chinese growth widened high-yield spreads worldwide; and in April 2020 (134 basis points), in the first weeks of the COVID pandemic. The February 2016 episode is the least familiar

of the three, and it is worth noting that it is not an artefact of our construction: Favara et al. (2016) document that over the same months the U.S. excess bond premium of Gilchrist and Zakrajšek (2012) rose to levels last seen in the early summer of 2008. That the series should rise at each of these dates is reassuring rather than surprising, but it is a necessary condition for a measure of global credit-market sentiment to be credible. During periods of elevated GEBP, dispersion across countries increases too, as indicated by the widening of the spread of the grey lines. The 5th–95th percentile range of the country EBPs, for example, averages 921 basis points between September 2008 and June 2009, against 191 basis points over the rest of the sample.

Neither the level nor the dynamics of the global series depend materially on the choices made in constructing it. Two of those choices are substantive. The first is whether the pricing regression (1) is estimated pooled across countries, as in the baseline, or country by country, so that every slope is free to differ. The second is the unit over which bond-level premia are averaged: equation (4) averages across bonds, which weights the U.S. heavily because it dominates global issuance, whereas averaging the country EBPs gives every country equal weight. Figure C.1 in Appendix C plots the baseline against both alternatives. Relaxing the common-slope restriction is all but immaterial: the country-by-country global series has a correlation of 0.997 with the baseline in levels and 0.999 in first differences, and departs from it by 4 basis points on average and by at most 12. Equal-country weighting matters more, as it must, but not qualitatively: that series correlates 0.96 with the baseline in levels and 0.93 in first differences, has a mean absolute gap of 14 basis points, and turns at the same dates with a very similar amplitude (a standard deviation of 66 basis points, against 67 for the baseline). We therefore work throughout with the pooled, bond-weighted series of equation (4).

Beneath the global series, however, lies considerable heterogeneity. Figure 3 plots every country EBP alongside the global series, and two things are apparent. The country series move together in the crisis episodes, all widening around the same dates, which is what makes a common factor plausible in the first place. Yet they differ markedly in amplitude: some countries barely move in episodes that push others out by several hundred basis points, and the dispersion across countries widens precisely when the global series spikes. Whether this heterogeneity is second-order relative to a dominant common component, or whether it is substantial enough that country-level series carry information of their own, is the question the next two subsections take up in turn.

3.2 One Common Factor

Figure 3 suggests strong comovement across countries. To establish this formally we estimate a static factor model on the panel of country EBPs. The panel is unbalanced: of the 41 countries, 14 are observed over the whole of January 1998 to June 2023, while the remainder for the most part enter late rather than being intermittently observed, so that missingness is predominantly a ragged-edge problem. We therefore estimate the factors by the EM principal-components algorithm of Stock and Watson (2002), which iterates between extracting principal components from the completed data matrix and replacing the missing entries by their fitted common component. The panel is the one constructed in Appendix B.2: each country's sample is dated from the end of any false start, and a country must have at least 120 monthly observations and within-span coverage of at least 80 percent, so that the estimate is not driven by imputation in series that are largely unobserved.[9]

Table 3 reports the variance decomposition over three nested samples: the balanced subpanel of 14 countries observed throughout, where principal components are exact and nothing is imputed (columns 1 and 2); the 30 countries whose effective sample is a single continuous spell, so that the algorithm has only ragged edges to fill in (column 3); and all 41 countries of the analysis panel (column 4). For the two unbalanced columns the statistic is the share of the variance of the *observed* entries explained by the common component, computed on those entries alone because the imputed ones are by construction exactly rank one.

The first factor accounts for the bulk of the variation on every sample: 77 percent of the variance of the balanced subpanel, 74 percent of the continuous-spell panel, and 68 percent of the full panel, with all 14 balanced countries loading on it with the same sign. Subsequent factors matter much less: in the balanced subpanel the second adds a further 6 percentage points and the third 5, so that five factors together reach 94 percent. This second claim rests on the balanced subpanel alone, and necessarily so: the EM iteration is a contraction at one factor but not beyond it, so with two or more factors the imputed cells diverge and

[9]The false-start rule affects 4 countries – Argentina, Ireland, Luxembourg, and Poland – each of which goes from a badly interrupted series to an uninterrupted one, at a cost of between 23 and 42 early observations. Given the issuer and spell screens already imposed on the bond sample, the length and coverage requirements turn out not to bind: all 41 countries satisfy them. Appendix Table B.2 documents coverage country by country.

Table 3 Variance of country EBPs explained by the common factors

	Balanced subpanel (exact PCA)		Unbalanced panel (EM-PCA)	
Factors	**Share**	**Cumulative**	**Continuous spells**	**All countries**
1	0.767	0.767	0.740	0.685
2	0.065	0.832	–	–
3	0.055	0.887	–	–
4	0.034	0.921	–	–
5	0.023	0.943	–	–
Countries		14	30	41

Note. Share of the variance of the country EBP panel accounted for by the leading common factors, sample January 1998 to June 2023. Columns 1 and 2 use the balanced subpanel of countries observed over the full window, where principal components are exact and no imputation is involved; they are the only columns on which the second and higher factors are identified. Columns 3 and 4 report the share of the variance of the *observed* entries explained by the common component of an EM principal-components decomposition of the unbalanced panel, following Stock and Watson (2002). Column 3 restricts the panel to countries whose effective sample is a single continuous spell, that is, whose longest interior gap is zero; column 4 uses every country in the analysis panel. Only the one-factor row is reported for these two columns: the EM iteration is a contraction at one factor but diverges beyond it, so the higher-order unbalanced shares are not identified. The three samples are nested. The analysis panel retains countries with at least 120 monthly observations and within-span coverage of at least 80 percent, criteria that in the event exclude no country; see Appendix Table B.2.

the higher-order shares of the unbalanced panel are not identified.[10]

Nor is the column-4 share an artefact of the imputation. Regressing the country EBP on country and time fixed effects, estimated from observed country-months only, the time effects account for 51 percent of the within-country variation, against 68 percent in column 4. The direction of the gap is informative: a time effect constrains every country to load on the common component identically, whereas the factor model lets loadings differ – and, as we show next, they differ a great deal. The fixed-effects share is therefore a conservative floor on how much of the panel is common, and it is already above one half. The two common components are, moreover, all but the same series, correlating at 0.99.

The estimated factor is also close to indistinguishable from a simple cross-country average, correlating with the unweighted mean of the country EBPs at 0.99 in levels and 0.98 in first differences, and with GEBP at 0.97 and 0.95. Nothing that follows therefore hinges on the factor-analytic machinery, and we retain GEBP as our headline global series: it is transparent, and it is directly comparable to the construction in Gilchrist and Zakrajšek

[10] At one factor the iteration converges in fewer than two hundred passes and the completed matrix stays well scaled, with a trace close to the number of series and imputed values inside the range of the observed data. At two factors the imputed cells grow to roughly sixteen times the largest observed value. All multi-factor evidence therefore comes from the balanced subpanel, which involves no imputation whatsoever.

(2012).

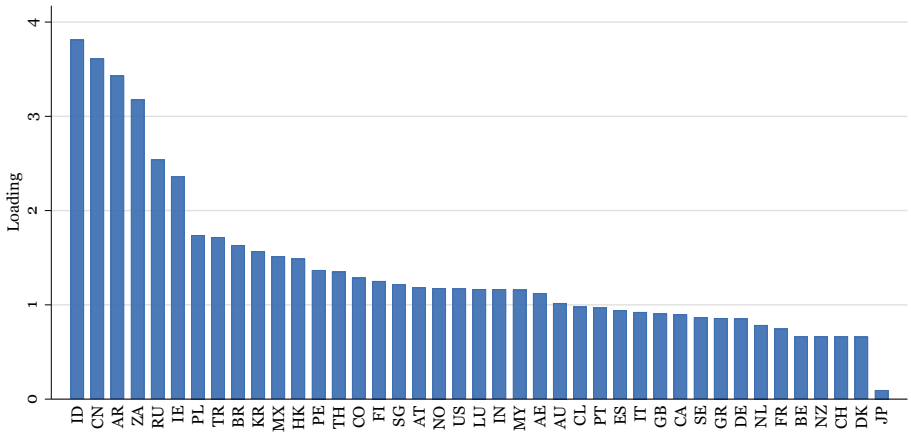
This is the paper's first main message: corporate credit conditions are, to a first approximation, a global phenomenon rather than a collection of national ones, as the literature on the global financial cycle would lead one to expect (see, e.g., Rey (2015) and Miranda-Agrippino and Rey (2020)). Common, however, does not mean uniform. Figure 4 reports, for every country in the analysis panel, two objects internal to the column-4 decomposition: the loading on the first common factor, in basis points of the country EBP per basis point of the global factor, which measures amplification; and the share of the country's observed EBP variance that the factor explains, which measures synchronization. The two orderings are far from identical, and the distinction between them is informative.

Advanced economies with large and liquid bond markets are highly synchronized but amplify comparatively little: the U.K. has the highest variance share in the sample, 0.96, with a loading well below unity (0.91), and the Netherlands, France, Germany, Belgium, and Australia display the same configuration. Emerging markets show the opposite pattern. Indonesia, China, Argentina, South Africa, Russia, and Turkey carry the largest loadings, from 1.72 to 3.81, but their variance shares are on average lower, so that global shocks reach them heavily amplified while a substantial share of their EBP variation remains domestic. Ireland is the one advanced economy that shares the amplification without the detachment, combining a loading of 2.36 with a variance share of 0.80, and financial centers sit in between: the loadings for Hong Kong and Luxembourg are 1.49 and 1.17. The U.S., consistent with its centrality in the global corporate bond market, combines a loading close to unity (1.17) with a high variance share (0.88); its EBP is also more volatile than that of any of the five next-largest issuers, with a standard deviation of 86 basis points against 10 to 65 for Japan, Canada, the U.K., Germany, and France. This may reflect greater heterogeneity among issuing firms: while the financial systems of the euro area and Japan are bank dominated, bond issuance in the U.S. is substantially more common for smaller and riskier firms.

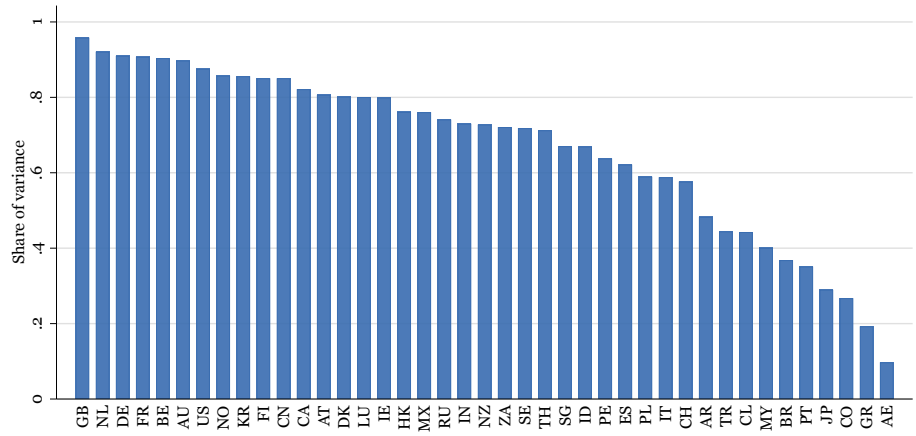
Japan is the outlier. Its loading, at 0.10, is the smallest in the sample and less than a fifth that of the next-lowest country, confirming the impression from Figure 3 that its EBP lies flat around zero for most of the sample period: Japanese corporate bond spreads are almost fully accounted for by default probabilities, leaving little scope for a risk-appetite component that might comove with anything else. Greece and the United Arab Emirates are disconnected in a different way, their loadings being close to unity, at 0.86 and 1.12, while their variance shares, 0.19 and 0.10, are the lowest in the sample.

Figure 4 Country Loadings on the Global Factor and Variance Explained

Panel A: Loading on the first common factor



Panel B: Share of variance explained



Note. Country loading on the first common factor (upper panel) and share of the country's observed EBP variance explained by that factor (lower panel), both taken from the EM principal-components decomposition of the unbalanced panel reported in column 4 of Table 3. The loading is expressed in basis points of the country EBP per basis point of the global factor, the factor having been rescaled to match the mean and standard deviation of GEBP. The variance share is computed over each country's observed months only. Sample January 1998 to June 2023, 41 countries. Countries are ordered from the largest to the smallest value in each panel.

Reading these two objects off the decomposition itself, rather than off country-by-country regressions on the estimated factor, is a matter of internal consistency rather than of substance: Appendix D reports the regression-based counterpart of Figure 4, whose slopes correlate with the loadings at 0.999 across countries and whose R^2 values correlate with the variance shares at 0.998.

3.3 Why Local Conditions Still Matter

The dominance of the global factor raises the question of whether the country-specific component is economically meaningful or merely measurement noise. Our results suggest it is not noise: the local EBP helps forecast domestic real activity over and above global and alternative domestic predictors, which is precisely what gives the local component its economic content.

The point is stronger than it may first appear. One might expect the local component to matter only where the common factor leaves a great deal unexplained – in Greece or the United Arab Emirates, say, whose variance shares in Figure 4 are close to zero. That is not what we find. The forecasting evidence below covers economies at the top of the synchronization ranking, including the U.K., Germany, France, and the U.S., whose variance shares range from 0.88 to 0.96. Even for these countries, where the global factor accounts for the bulk of the variation in the local EBP, the residual fraction carries predictive content for domestic activity. Both components therefore matter: the global factor dominates the variance, but what it leaves behind is economically informative rather than residual noise.

We establish this by replicating the forecasting exercise (documented for the U.S. in Appendix E) for other major economies in our sample. As indicators of macroeconomic activity, we again include the unemployment rate and industrial production in the manufacturing sector. Across economies, the country-EBP is the dominant factor in predicting monthly economic activity (Tables F.4–F.12). A higher country-EBP is a consistent predictor for a higher unemployment rate and lower industrial production in the next quarter and year, with the predictive power being comparable over the two horizons. Additional forecasting exercises are reported in Appendix F, Tables F.2 to F.14. These forecasting exercises suggest that country-level EBPs should be included to improve inference in forecasting exercises of local economic activity, as well as in VAR- or LP-based analyses of economic policies.[11]

[11] As noted by Montiel Olea et al. (2026), researchers are generally advised to include variables that are strong predictors of the outcome as controls.

Table 4 Forecasting exercise - 12-month ahead unemployment rate

	(1) AU	(2) CA	(3) DE	(4) FR	(5) GB	(6) JP	(7) US
Term spread	-0.42***	-0.26	-0.30***	0.18**	-0.25***	0.29***	-0.23
	(0.10)	(0.24)	(0.06)	(0.09)	(0.09)	(0.10)	(0.17)
Real policy rate	0.00	0.45**	-0.08	0.22*	0.03	-0.18*	0.46**
	(0.12)	(0.20)	(0.06)	(0.12)	(0.10)	(0.10)	(0.18)
Predicted Spread	0.40**	-0.40*	-0.60**	1.14***	0.27	0.46	0.39*
	(0.20)	(0.22)	(0.25)	(0.33)	(0.29)	(0.49)	(0.23)
EBP	0.14***	0.56***	0.17***	0.38***	0.24***	1.08***	0.51***
	(0.05)	(0.09)	(0.04)	(0.08)	(0.08)	(0.25)	(0.10)
R-squared	0.413	0.297	0.464	0.546	0.376	0.240	0.336
Obs.	282	282	174	222	249	290	282

Note. Columns are countries; the dependent variable is the change in the unemployment rate over the following 12 months. The excess bond premium enters as the country EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

4 The Role of Monetary Policy in Global Credit Conditions

Having established the importance of the country-specific EBPs for forecasting local economic activity and confirmed that our country-level EBPs are well behaved, we now turn to the role of monetary policy in shaping the various EBPs. It is well established that the U.S. EBP is highly responsive to U.S. monetary policy (Gertler and Karadi (2015), Anderson and Cesa-Bianchi (2024)). To assess the effect of monetary policy surprises on EBPs for a wider range of countries, we combine the EBP data with high-frequency monetary policy shocks for three economies. In particular, we rely on the surprises by Bauer and Swanson (2023) for the U.S., Jarociński and Karadi (2020) for the euro area, and Braun et al. (2025) for the U.K.

We estimate local projections following Jordà (2005) for the GEBP (and analogously for the Euro Area EBP) as well as on the panel of country-specific EBP using the following

regression specifications:

$$GEBP_{t+h} = \alpha^h + \beta_1^h MPS_{k,t} + \sum_{l=1}^{L} \beta_2^h GEBP_{t-l} + \epsilon_{t+h} \tag{6}$$

$$EBP_{k,t+h} = \alpha^h + \beta_1^h MPS_{k,t} + \sum_{l=1}^{L} \beta_2^h EBP_{k,t-l} + \sum_{l=1}^{L} \beta_3^h X_{k,t-l} + \gamma_k^h + \epsilon_{k,t+h} \tag{7}$$

Controls X for country k include its GDP, CPI, the nominal exchange rate, and equity prices (all logarithmically-transformed). As our baseline, we select a lag-length of $L = 12$ months and our estimation sample is restricted to end in December 2019, before the outbreak of the COVID-19 pandemic. In addition, we estimate country-specific local projections for each EBP series separately:

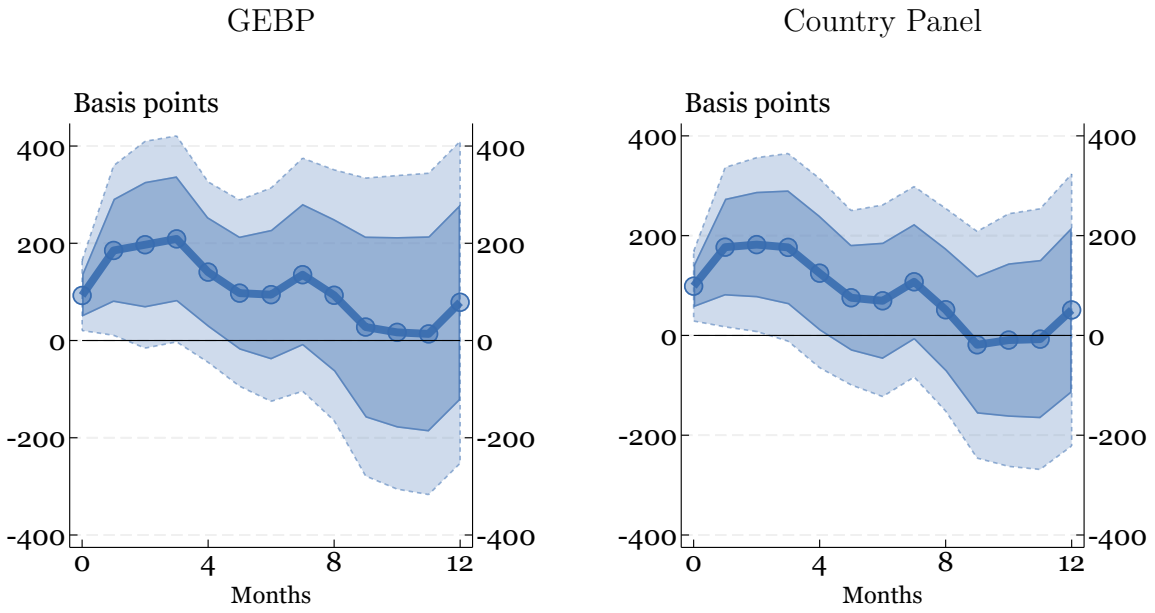
$$EBP_{k,t+h} = \alpha_k^h + \beta_{1,k}^h MPS_{k,t} + \sum_{l=1}^{L} \beta_{2,k}^h EBP_{k,t-l} + \sum_{l=1}^{L} \beta_{3,k}^h X_{k,t-l} + \epsilon_{k,t+h} \tag{8}$$

We restrict the sample for this country-level regression to countries for which we have a continuous spell of at least 200 months, which reduces the sample to span 20 countries. Both models are estimated separately for the three different monetary policy shocks.

Figure 5 documents our first key finding: a one-standard deviation contractionary U.S. monetary policy shock is associated with an increase in the GEBP series of up to 200 basis points after one month. The effect reaches its maximum strength after one to three months and dies out completely after around eight months. The observation that the effect of monetary policy on the GEBP is short-lived is in line with the dynamics documented by Gertler and Karadi (2015) for the U.S. EBP. These results are confirmed if we instead estimate the effect of U.S. monetary policy shocks on the country panel of EBP. Again, we find a similar magnitude of the peak effect and comparable dynamics.

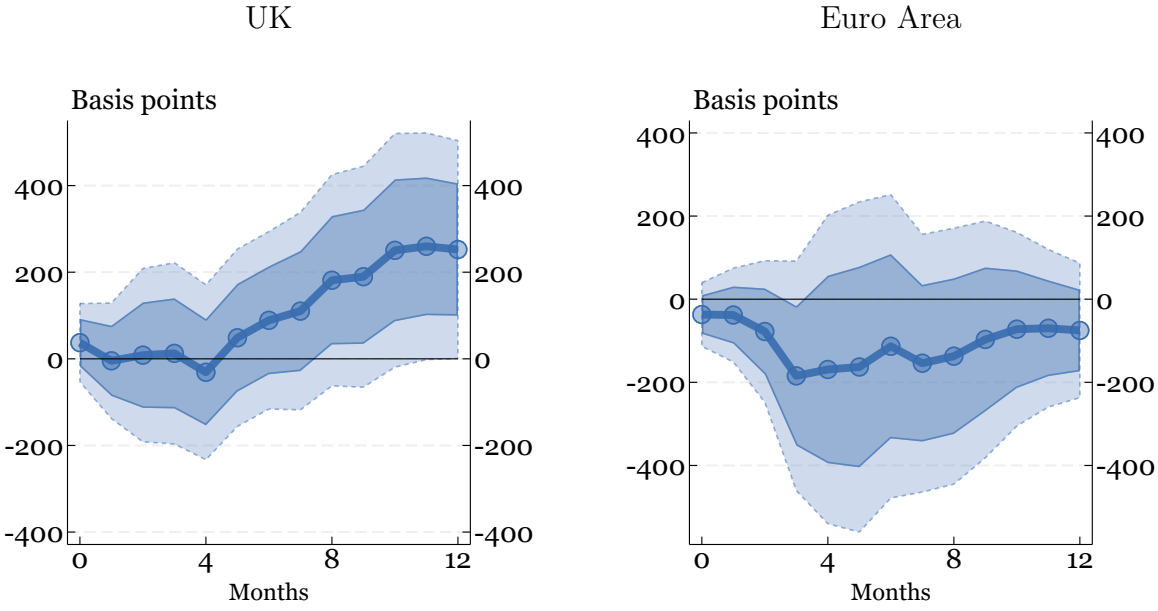
Among other central banks, we would expect monetary decisions by the European Central Bank or the Bank of England to be the most likely to affect global financial conditions. Interestingly, however, none of the other monetary policy shocks seem to affect global financial conditions as measured by the GEBP (Figure 6). This finding is in line with evidence of a "global financial cycle" (Rey (2015), Miranda-Agrippino and Rey (2020)) driven by U.S. monetary policy. These empirical results are also supportive of theoretical work by Kekre and Lenel (2025), which derives a global financial cycle driven by U.S. monetary policy in a model of international arbitrageurs funded by the U.S. dollar.

Figure 5 Response of EBP worldwide to US Monetary Policy shocks



Notes: Dynamic response of the EBP worldwide to US monetary policy shocks. Shaded areas correspond to the 68% and 90% confidence intervals, where heteroskedasticity-robust standard errors are used for the GEBP specification and two-way clustering at the country and month level for the panel regression.

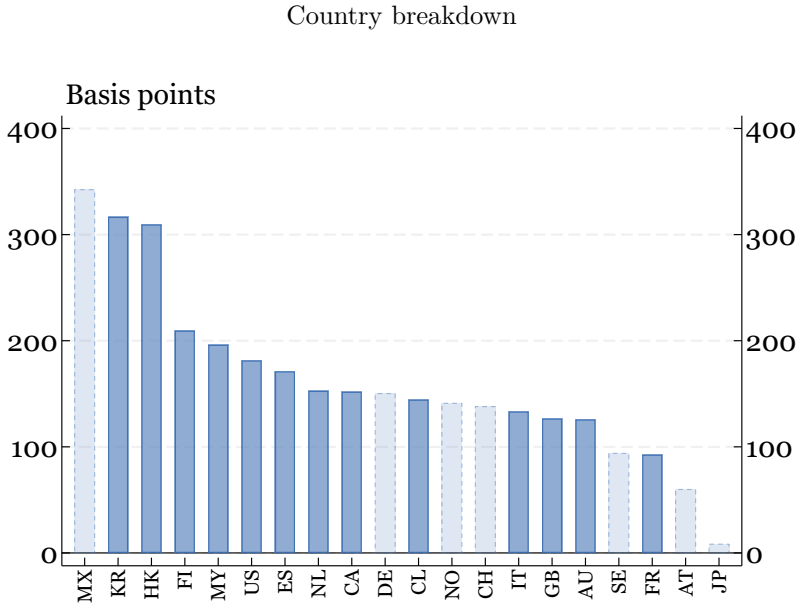
Figure 6 Response of the GEBP to U.K. and Euro Area Monetary Policy Shocks



Notes: Dynamic response of the GEBP to US, UK, and euro area monetary policy shocks. Shaded areas correspond to the 68% and 90% confidence intervals, using heteroskedasticity-robust standard errors.

When considering cross-country variation in the effect of the U.S. monetary policy shock on the EBP, we find a significant positive effect for a majority of countries and a positive effect for all countries with sufficient continuous EBP observations (Figure 7).

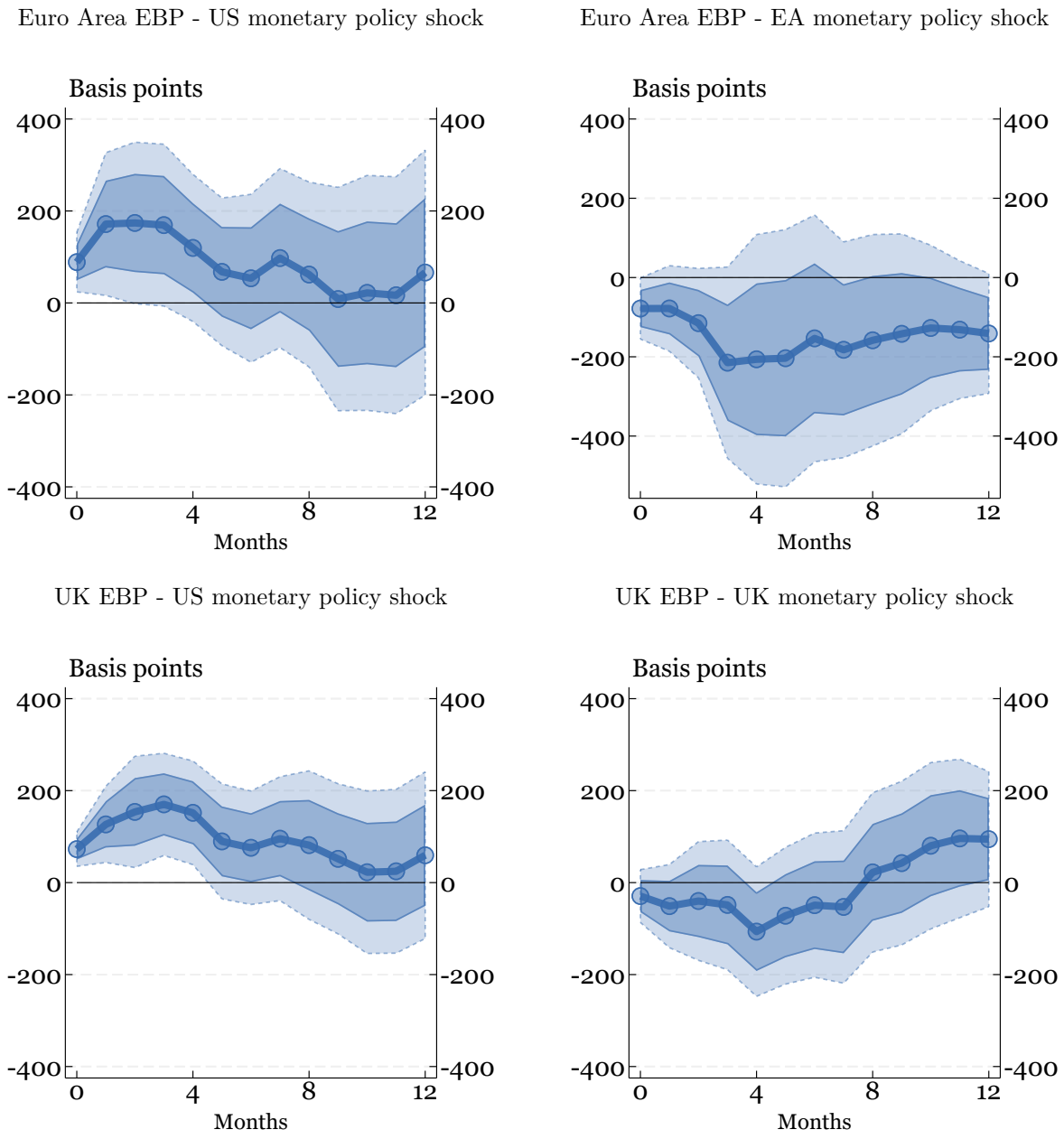
Figure 7 Country-Level Responses of the EBP to U.S. Monetary Policy Shocks



Notes: Country-specific responses to the US monetary policy shock calculated in a country-level regression for a one-month horizon. Filled bars indicate significance of the coefficient on a 10% significance level using heteroskedasticity-robust standard errors.

Strikingly, Figure 8 shows that euro area and U.K. monetary policy surprises do not even affect financing conditions on the domestic corporate bond market. That is, a U.K. monetary policy shock does not have a significant effect on the U.K. EBP, while a euro area monetary policy shock does not have a significant effect on the euro area EBP. This finding is not specific to the monetary policy shock series chosen (see Appendix, Figure H.2). Both EBP series are, however, responsive to U.S. monetary policy surprises.

Figure 8 Response of the Euro Area and U.K. EBP to U.S. versus Domestic Monetary Policy Shocks



Notes: Dynamic response of Euro Area EBP (top panel) and UK EBP (bottom panel) to a US monetary policy shock (left panel) and a local monetary policy shock (right panel). Shaded areas correspond to the 68% and 90% confidence interval with heteroskedasticity-robust standard errors.

5 Tracing the Transmission Channel

When it comes to the sensitivity of financial conditions across the globe to U.S. monetary policy, and *U.S.* monetary policy only, explanations can either be borrower- or lender-based. Broadly speaking, the potential channels can be summarized as follows:

- *Borrower-based*: a Fed tightening slows down the world economy (due to the sheer importance of U.S. demand) and/or pushes up financing costs for many firms globally (given the prevalence of USD-denominated borrowing). Both factors can increase corporate default risk (especially for USD-indebted firms whose revenues are primarily denominated in some other currency), making firms around the globe riskier to lend to. In this case, the rise in corporate financing costs reflects the borrower's reduced creditworthiness.

- *Lender-based*: a Fed tightening lowers the risk-bearing capacity of financial intermediaries. This increases the "price of risk" they will charge per unit of risk they take on, e.g. via bonded lending to corporations. In this case, any "global financial cycle" should be particularly visible in EBPs associated with riskier borrowers (without the Fed tightening directly affecting borrower creditworthiness, i.e., the quantity of risk which lending to firms embeds). Lender-based explanations come in two variants, which differ in *where* the dollar matters. In the first, the cost attaches to the *bond*: a Fed tightening raises the cost of funding and hedging any USD-denominated asset, so every dollar bond should reprice, whoever issued it (Avdjiev et al. (2019)). In the second, the cost attaches to the *holder*: USD bonds of non-U.S. issuers are disproportionately held and intermediated by non-U.S. institutions, whose wholesale dollar funding is particularly sensitive to U.S. monetary policy (Ivashina et al. (2015)), so only the dollar paper on their balance sheets should reprice.

In this section, we will argue that our results point to *lender-based* explanations, and within those to the second variant: the dollar matters through who holds the bond, not through the currency the bond is written in. This is consistent with the theoretical mechanisms proposed in papers such as Avdjiev et al. (2019), Bräuning and Ivashina (2020), and Kekre and Lenel (2025), and results on international capital flows documented in Bruno and Shin (2015a).

5.1 Borrower- versus lender-based explanations

As set out above, borrower- versus lender-based explanations for U.S.-driven changes in global financial conditions differ in where they predict the spillovers to show up. According to borrower-based theories, they should emerge in borrower creditworthiness; lender-based theories predict them to show up in the residual, EBP component (which is thought to proxy for lenders' risk-bearing capacity – and the price of risk they charge).

While we have previously shown how U.S. monetary policy shocks affect EBPs around the globe – which is pointing towards a lender-based explanation – it is possible that any observed movements in EBPs are secondary to *even larger* movements in the predicted component of the spread, $\hat{S}$, reflecting default risk. Figure 9 looks into this possibility, by comparing the EBP responses to responses in $\hat{S}$ for the major countries in our panel (and recall that these sum up to responses in observed bond spreads by definition of the EBP: $S_t = \hat{S}_t + EBP_t$, see Equation 3).

As the figure shows, there is no country for which the predicted spread displays a significant positive response, and the point estimates are small relative to those of the EBP.[12] This suggests that the influence of U.S. monetary policy over global financial conditions is reflecting a lender-side phenomenon.

5.2 Further evidence at a more granular level

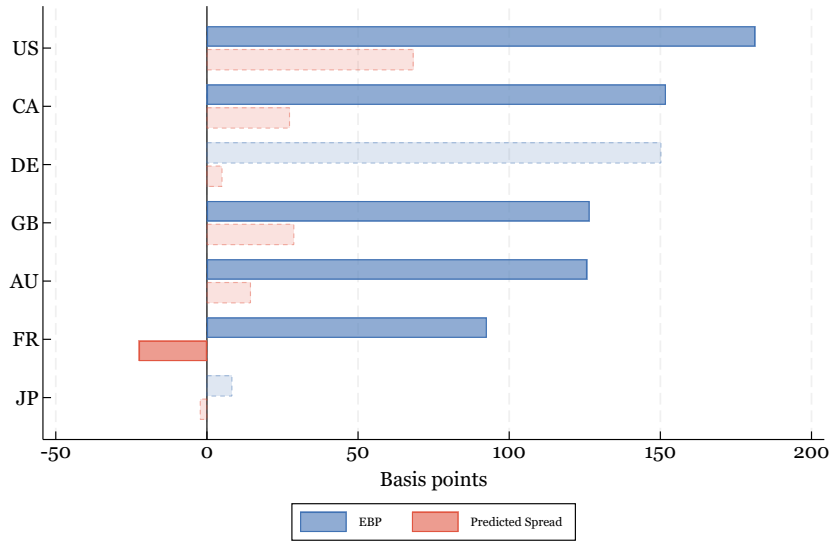
Distinguishing between borrower- and lender-based explanations can also be done by exploiting our data at a more granular level. There we can see how firms' default risk changes in response to a U.S. monetary policy shock, and how any given amount of risk is priced. In particular, our underlying dataset (from which our country-level EBPs are constructed) contains bond-level observations, indexed by i, with us observing the identity of the issuing firms, indexed by j. This enables us to run regressions of the following type at the bond level:

$$y_{i,t+h} = \alpha_i + \lambda_{c(j),s(j),t} + \gamma(\varepsilon_t^{US} \times x_{j,t-1}) + \delta x_{j,t-1} + \sum_{l=1}^{12} \rho_l y_{i,t-l} + e_{i,t+h}. \tag{9}$$

Here, $y_{i,t+h}$ is the outcome variable of interest, for which we alternatively consider (a)

[12] The figure focuses on the $h = 1$ horizon, but other horizons don't show a significant impact either.

Figure 9 Response of the EBP versus Predicted Spread to a U.S. Monetary Policy Shock across selected countries (h=1)



Notes: Country-specific responses to the US monetary policy shock calculated in a country-level regression for a one-month horizon.

the total spread (S), (b) its fitted component ($\hat{S}$, which is the spread predicted from the underlying default probability and other covariates; recall Equation 1), and (c) the EBP ($\equiv S - \hat{S}$). α_i is a bond fixed effect (and since a bond is issued by one firm only, it consists of a firm fixed effect plus a bond-specific fixed effect), while $\lambda_{c(j),s(j),t}$ is a country $\times$ sector $\times$ month effect; $x_{j,t-1}$is a firm characteristic, where we consider three measures of firm riskiness: (i) a high-yield dummy, (ii) the log default probability, and (iii) the dollar share of the firm's bonded debt. In the latter case, we limit our sample to issuers incorporated *outside* of the U.S., making it more likely that these firms carry some type of currency mismatch (having USD-denominated debt, but relying on a significant share of non-USD revenues) increasing their riskiness along that dimension.

In this setup, γ is our coefficient of interest. It tells us what firm-level characteristics make the outcome variable y particularly sensitive to a U.S. monetary policy shock, ε_t^{US}.[13] Results – always for $h = 1$ month out (which is where we typically find the EBP, or spread, response to peak in aggregate) – are summarized in Table 5. They lend further support to the notion that our findings reflect a lender-side pattern. In particular, Table 5 shows that spreads of high-yield firms are more sensitive to U.S. monetary policy shocks – and that all

[13]Interacting the euro area and U.K. shocks with the same firm characteristics yields coefficients of no consistent sign, which do not appear at other horizons, and which belong to shock series for which Section 4 finds no level effect on any EBP.

of this is coming from the EBP, *not* from the response of the firm's riskiness itself. The same holds for firms with a higher estimated probability of default, and for non-U.S. firms with a larger share of their bonded debt denominated in USD.

Table 5 Firm-level response to a US monetary policy shock by issuer credit risk

x	EBP	S	SF	Sample
log PD, per sd	4.1*** (1.6)	4.4*** (1.6)	0.3 (0.2)	589k
HY minus IG	10.9* (5.5)	12.1** (5.9)	1.5 (1.3)	589k
Dollar share of firm debt	7.7** (3.6)	7.7** (3.5)	−0.3 (0.5)	214k

Note. Interaction coefficient γ of Equation 9 for horizon $h = 1$. Columns represent outcome variables y, while rows correspond to the relevant interacted firm characteristics x. Standard errors are clustered at the firm and month level. *, ** and *** denote significance at the 10%, 5% and 1% levels.

Recognizing the corporate bond spread as a risk premium, which is the price of risk times the quantity of risk, these results suggest that it is *the price of risk* – as proxied by the EBP – which is pushing the spread up after a U.S. monetary policy contraction. A Fed tightening appears to make lenders more risk averse, which especially pushes up the borrowing costs of riskier firms (this is what $\gamma > 0$ implies). This seems to happen *without* these firms actually becoming any riskier relative to their pre-shock situation, as evidenced by the fitted component of the spread not showing a significant response.[14] From this, we conclude that we are observing a pattern that is primarily driven by lenders charging a higher *price of risk* following a U.S. monetary policy tightening; the quantity of risk (proxied by borrowing firms' creditworthiness) does not show a significant response.

A first indication that the currency of denomination is not the relevant margin comes from simple bond-level projections with bond fixed effects: the EBP on euro-denominated bonds of euro area issuers rises by 8.2 basis points per one-standard-deviation U.S. shock at $h = 1$, against 10.1 basis points for USD-denominated bonds of U.S. issuers, while neither the ECB nor the Bank of England moves the premium on bonds in its own currency.[15] If the dollar matters, it must therefore matter through a channel that an average across all dollar bonds conceals.

Exploiting the granularity of our dataset, we can dig a little deeper and see whether the residency of the borrowing firm matters, holding other factors (including currency of denomination) constant. Here, the underlying idea is that USD bonds issued by *non-U.S.*

[14] The same conclusion follows by looking at the response of the default probability itself.

[15] Standard errors are 3.8 and 4.9 basis points, respectively; untabulated.

corporations are disproportionately held and intermediated by non-U.S. financial institutions (i.e., that there is some "home bias"; Coeurdacier and Rey (2013)). The dollar funding and hedging costs of non-U.S. financial institutions have been shown to be particularly sensitive to U.S. monetary policy due to foreign banks' stronger reliance on wholesale funding as opposed to stickier dollar deposit funding (Ivashina et al. (2015)). This would make these non-U.S. financial institutions act in a more risk averse manner after a Fed tightening – predicting a greater EBP sensitivity of USD bonds issued by non-U.S. entities to U.S. monetary policy.

To examine this possibility, we estimate regressions of the following form (on a sample that has dropped offshore incorporations, as it is not clear in which direction their "home bias" should run):

$$\begin{aligned} EBP_{i,t+h} =& \alpha_{j,t} + \beta_1(\varepsilon_t^{US} \times USD_i) + \beta_2(\varepsilon_t^{US} \times USD_i \times nonUS_j) \\ &+ \gamma_1 USD_i + \gamma_2(USD_i \times nonUS_j) + \sum_{l=1}^{12} \rho_l EBP_{i,t-l} + e_{i,t+h}. \end{aligned} \tag{10}$$

Here, USD_i is a dummy variable that takes on the value 1 when bond i is issued in USD (and 0 otherwise), while $nonUS_j$ takes the value 1 when the issuing firm j is incorporated outside of the U.S. Note that equation (10) does not feature macro controls (such as the U.S. monetary policy shock, ε_t^{US}, by itself) as they are collinear with the issuer-month fixed effects, $\alpha_{j,t}$.

With this regression we ask: within a single firm, do its USD bonds respond differently to a U.S. monetary policy shock (compared to "sister bonds" issued in a different currency) and does the answer to this question depend on whether the firm is located inside or outside the U.S. (the idea being that firms located outside of the U.S. are more reliant on non-U.S. investors)?

In particular, β_1 measures the within-firm difference for U.S.-incorporated firms between the response of the EBP on bonds issued in USD, relative to bonds issued in other currencies; $\beta_1 + \beta_2$ captures the same gap for issuers based outside the U.S., with β_2 thus capturing the difference between them.

Results of this exercise are shown in Table 6. The within-firm gap is 3.7 basis points for non-U.S. issuers and 0.8 basis points for U.S. issuers; the latter is not significantly different from zero, while the difference between the two, β_2, is positive and significant at the 5% level. This suggests that the response of the EBP to a U.S. monetary policy shock is not a strong reflection of the bond's currency, but rather of the bond's holder: the EBPs of

USD bonds issued by non-U.S. corporations (which are disproportionately held by non-U.S. investors) are most sensitive to U.S. monetary policy shocks. Our result for β_1 implies that, when solely looking at U.S.-based corporations, the EBPs of USD-denominated bonds are *not* more sensitive to U.S. monetary policy (compared to bonds issued by the same firm in other currencies). This discriminates among lender-based mechanisms: a funding cost that attached to the bond's currency would not know the issuer's residency, and would move a U.S. firm's dollar bond against its own non-dollar bonds by as much as it moves a non-U.S. firm's. It does not. The pattern is instead the one predicted when the cost attaches to the holder, with non-U.S. intermediaries repricing the dollar paper they hold and make markets in.

Table 6 Within-firm currency gap in response to a US monetary policy shock for US and non-US issuers

	Within-firm currency gap
Non-US issuers, $\beta_1 + \beta_2$	3.7* (2.2)
US issuers, β_1	0.8 (1.9)
Difference (non-US − US), β_2	2.9** (1.3)
Pooled dollar dummy, no issuer split	2.6 (2.0)

Note. Coefficients of equation (10) giving the within-firm currency gap for US issuers, non-US issuers, and the difference between them following a US monetary policy shock. Standard errors are clustered at the firm and month level. *, ** and *** denote significance at the 10%, 5% and 1% levels.

Read together, the results of this section say what U.S. monetary policy moves. First, and largest, a single price of credit risk, applied to every bond in the panel whatever its currency or its issuer's residency, scaled by the bond's quantity of risk (Table 5), and absent from the default component of the spread. A marginal lender pricing risk off a single state variable, its risk-bearing capacity, across the whole global corporate bond market is also why Section 3.2 finds a single common factor in country EBPs. Second, and smaller, a surcharge of three to four basis points at horizons of one to two months on the dollar paper of non-U.S. issuers, absent from U.S. issuers' dollar paper (Table 6): the signature of non-U.S. intermediaries whose dollar funding the Fed reaches. Domestic monetary policy shocks move neither.

6 Conclusion

This paper introduces and analyzes a novel cross-country dataset of Excess Bond Premiums (EBPs), constructed using a harmonized methodology for 41 countries and validated against existing alternative measures. Our forecasting exercises demonstrate that country-level EBPs contain valuable information for predicting future economic activity and improve predictive performance when incorporated into forecasting models. We further show that EBPs across countries are driven primarily by U.S. monetary policy, whereas monetary policy surprises originating from other central banks have no statistically significant effect on credit risk premia. These findings provide strong support for the "Global Financial Cycle" hypothesis, whereby global credit market conditions are shaped by U.S. monetary policy through its effects on the funding costs of globally active financial intermediaries. Nevertheless, country-specific financial conditions retain significant predictive power for future economic activity, underscoring the importance of accounting for local variation alongside the dominant global component. Our dataset provides a new resource for researchers and policymakers seeking to understand the global determinants of corporate credit market conditions and their implications for economic performance.

This study, and its accompanying data, opens up new directions for research on global financing conditions: The provided data can help to answer questions around the role of global capital flows, of financial regulation, or macroeconomic policies in shaping credit conditions on corporate bond markets worldwide. The data might also prove useful in assessing global integration of financial markets.

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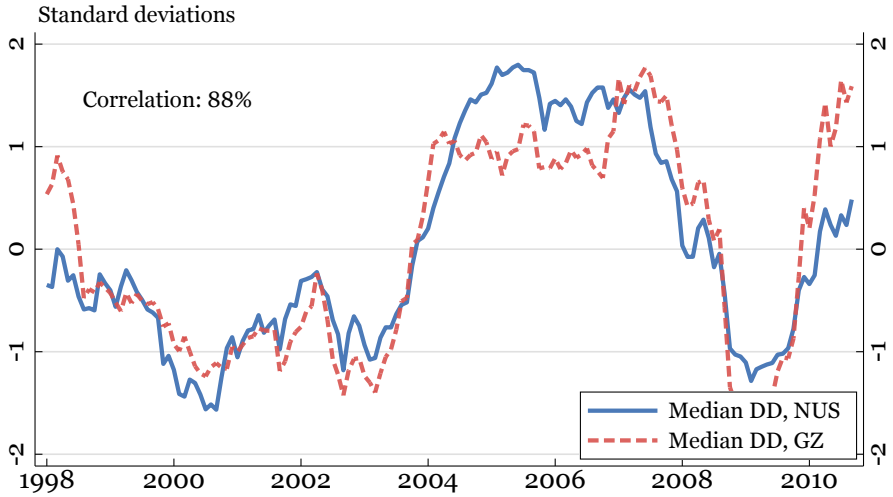
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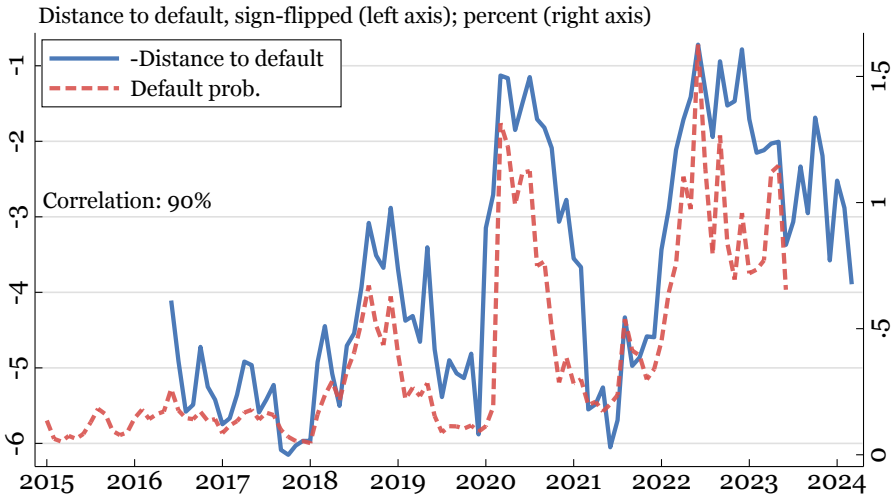
Appendix A: Data

Figure A.1 US Median Distance-to-Default: NUS-CRI versus GZ



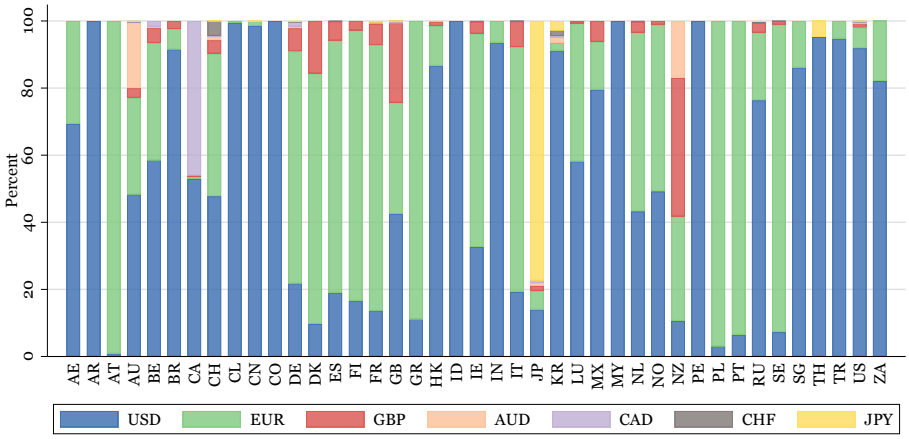
Note. Median distance-to-default of US firms implied by the NUS-CRI 12-month default probability (solid black), against the Merton distance-to-default of Gilchrist and Zakrajšek (2012) (dashed red). Face-value weighted, overlapping sample, January 1998 to September 2010. Both series are standardized to have mean zero and unitary variance.

Figure A.2 General Motors: PD versus Distance-to-Default



Note. The negative of the Merton (1974) distance to default (solid, left axis) against the NUS-CRI 12-month default probability (dashed, right axis) for General Motors, monthly from January 2015; the distance to default is available from June 2016, and the default probability is averaged across the firm's bonds within each month. The sign of the distance to default is flipped so that both series rise with default risk. The on-figure annotation reports the correlation between the two series over their overlapping sample. Sources: NUS-CRI default probabilities, Compustat and LSEG market data.

Figure A.3 Currency Composition of Corporate Debt, by Country



Note. Currency composition of corporate bond issuance by country, computed as each currency's share of the cumulative USD face value issued by the country's firms in the estimation sample of the pricing regression. Bars are stacked in the order USD, EUR, GBP, AUD, CAD, CHF, JPY, and the seven currencies exhaust the sample: no bond outside them passes the filters of Appendix B.1. Pre-euro national currencies are mapped to the euro. Source: LSEG/BAML bond data.

Table A.1 Monthly Count of Distinct Firms and Bonds, by Country

	Firms			Bonds				Firms			Bonds		
Country	Mean	Q1	Q3	Mean	Q1	Q3	Country	Mean	Q1	Q3	Mean	Q1	Q3
AE	4.4	2.0	6.0	7.3	2.0	11.0	IN	9.4	4.0	15.0	25.2	6.0	43.0
AR	5.8	5.0	7.0	9.9	7.0	13.0	IT	11.7	9.0	16.0	76.8	38.0	113.0
AT	3.6	2.0	5.0	9.5	3.0	14.0	JP	53.1	36.0	71.0	392.3	228.0	531.0
AU	24.7	19.0	35.0	64.3	42.0	99.0	KR	12.7	7.0	20.0	33.0	13.0	50.5
BE	4.9	3.0	6.0	32.8	9.0	55.0	LU	2.9	2.0	4.0	6.0	3.0	9.0
BR	16.9	17.0	23.0	53.5	27.5	79.5	MX	13.8	6.0	20.0	37.2	11.0	58.0
CA	66.3	48.0	82.0	347.3	222.0	492.0	MY	3.4	3.0	4.0	5.5	4.0	7.0
CH	9.2	4.0	13.0	54.5	10.0	89.0	NL	13.9	11.0	16.0	60.7	40.0	83.0
CL	6.9	3.0	11.0	11.3	6.0	17.0	NO	3.9	4.0	5.0	11.8	7.0	14.0
CN	43.1	19.0	64.0	91.9	21.0	136.0	NZ	1.9	1.0	3.0	3.7	2.0	5.0
CO	5.3	4.0	6.0	10.2	9.0	12.0	PE	5.5	5.0	7.0	9.4	8.0	12.0
DE	29.1	18.0	39.0	216.9	100.0	312.0	PL	3.1	1.0	4.0	3.9	2.0	5.0
DK	3.5	1.0	4.0	15.0	4.0	22.0	PT	2.9	2.0	3.0	9.6	6.0	13.0
ES	10.7	7.0	15.0	59.2	28.0	92.0	RU	9.4	7.0	12.0	40.4	37.0	46.0
FI	5.6	4.0	7.0	15.6	12.0	18.0	SE	9.8	8.0	10.0	21.0	14.0	23.0
FR	37.3	25.0	49.0	221.9	96.0	337.0	SG	1.9	1.0	3.0	4.1	3.0	6.0
GB	59.8	56.0	68.0	282.4	198.0	396.0	TH	3.8	2.0	5.0	8.1	5.0	11.0
GR	3.0	2.0	3.0	5.2	4.0	6.0	TR	8.0	8.0	9.0	10.6	10.0	12.0
HK	9.6	5.0	13.0	22.2	10.0	29.0	US	600.9	358.0	792.0	2915.9	1585.0	4327.0
ID	5.5	5.0	6.0	8.1	7.0	9.0	ZA	4.7	1.0	8.0	11.1	3.0	17.0
IE	3.7	2.0	5.0	13.6	5.0	24.0							

Note. Number of distinct firms and bonds observed per month, by country. For each country, columns report the mean and interquartile range (Q1, Q3) of the monthly count across months. Countries are listed alphabetically, read down the left panel and then down the right.

Appendix B: Sample construction and coverage

This appendix documents the two samples on which the analysis rests. The first is the bond-month panel on which the credit-spread pricing regression of Section 2 is estimated. The second is the panel of country EBP series that the analysis of Section 3 uses. They are distinct, and deliberately so. The pricing regression is a cross-sectional relationship between spreads and default risk, and is estimated on the widest bond panel that passes the data filters: a country that contributes few bond-months still contributes information about that relationship. A country EBP series, by contrast, is only interpretable as a time series if it is long enough and complete enough to be read as one. Restricting the pricing regression to the countries that satisfy the second requirement would discard usable pricing information for no gain, so we do not do it.

B.1 Filters applied to the bond sample

The following restrictions are imposed, in order, on the bond-month panel before the pricing regression is estimated. Each is applied uniformly to all countries.

1. *Sample period.* Bond-months before January 1998 are dropped. The earlier records are sparse and contain implausible outliers.
2. *Minimum country coverage.* Countries contributing fewer than 100 bond-month records in total are dropped.
3. *Instrument type.* Only senior bonds issued by non-financial corporations are retained.
4. *Face value.* Bond-months with a face value below USD 1 million are dropped.
5. *Spread range.* Bond-months with an option-adjusted spread below 5 basis points, as in Gilchrist and Zakrajšek (2012), or above 3,500 basis points, are dropped.
6. *Maturity.* Bond-months with less than one year or more than thirty years of remaining maturity are dropped.
7. *Matched default probability.* A bond-month is retained only if the issuer can be matched to an NUS-CRI twelve-month default probability.
8. *Country screens.* A country is retained only if it contains at least 8 distinct issuers over the whole sample and has an uninterrupted spread-and-default-probability spell of at least 36 months.
9. *Winsorization.* The option-adjusted spread and the log default probability are winsorized at the 1st and 99th percentiles, taken *within each month* and pooled across countries.

Two features of the last step are worth making explicit. First, the percentiles are taken within the month rather than within the country. The cap is therefore time-varying – loose in crises, tight in calm periods – so that it binds on issuers that are distressed relative to the rest of the world at that date, rather than relative to their own country's history. A country whose issuers are uniformly distressed has a high own 99th percentile, and winsorizing within the country would leave precisely the observations of interest uncapped. Winsorizing within the country-month is not a usable alternative: the cells that create the problem contain a handful of bonds, and with four observations the 99th percentile coincides with the maximum, so the cap never binds where it is needed.

Second, the winsorization is applied *after* the filters above rather than before them, so that the percentiles are computed on the sample the regression actually uses. The default-risk regressor is winsorized in logs rather than in levels. A non-negligible share of bond-months carries a default probability of exactly zero and hence no log; winsorizing the level would lift these to a strictly positive cap and thereby add them to the estimation sample, which would be a change of sample rather than a treatment of outliers.

Table B.1 reports the coverage of the resulting estimation sample country by country. The sample contains 1,469,405 bond-months, 3,931 issuers and 31,537 bonds across 41 countries.

Table B.1 Coverage of the Pricing-Regression Estimation Sample

Country	Obs.	Firms	Bonds	Sample	Cov.
US	825,388	2,284	19,300	1998m1–2023m6	100.0
JP	119,565	138	2,217	1998m1–2023m6	100.0
CA	89,342	293	1,614	1998m1–2023m6	100.0
GB	85,412	246	1,620	1998m1–2023m6	100.0
FR	67,018	116	1,192	1998m1–2023m6	100.0
DE	64,805	113	1,555	1998m1–2023m6	100.0
IT	21,015	39	345	2000m6–2023m6	100.0
AU	18,897	107	416	1998m1–2023m6	100.0
NL	18,336	78	410	1998m1–2023m6	100.0
ES	18,071	38	304	1998m1–2023m6	100.0
CN	17,056	147	554	2008m1–2023m6	100.0
CH	16,192	45	425	1998m1–2023m6	100.0
BR	12,624	58	234	2001m5–2023m6	88.7
MX	10,199	59	199	1998m1–2023m6	89.9
KR	9,565	37	244	1999m3–2023m6	100.0
BE	8,564	22	165	2001m4–2023m6	97.8
RU	7,613	18	124	2006m7–2022m3	100.0
HK	6,475	69	195	1998m1–2023m6	100.0
SE	6,312	35	150	1998m1–2023m6	100.0
IN	4,959	26	89	2007m2–2023m6	100.0

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Country	Obs.	Firms	Bonds	Sample	Cov.
FI	4,438	20	84	1999m10–2023m6	100.0
NO	3,566	14	76	1998m1–2023m6	100.0
CL	3,441	31	62	1998m1–2023m6	100.0
IE	2,834	14	69	2001m6–2023m6	79.2
AT	2,739	13	46	1998m12–2023m6	100.0
PT	2,733	9	49	1999m4–2023m6	98.3
ZA	2,566	12	59	2002m6–2023m6	91.7
DK	2,525	11	61	2009m5–2023m6	100.0
TH	1,830	11	32	2004m8–2023m6	100.0
AR	1,609	20	43	1998m1–2023m6	52.9
MY	1,607	11	22	1998m1–2023m6	95.1
LU	1,545	26	64	1998m1–2023m6	86.3
CO	1,483	13	32	2011m6–2023m6	100.0
GR	1,366	11	34	2000m2–2023m6	93.6
TR	1,351	10	22	2012m11–2023m6	100.0
PE	1,307	12	22	2011m12–2023m6	100.0
AE	1,230	9	24	2009m6–2023m6	99.4
ID	1,119	18	37	2010m11–2023m6	91.4
NZ	984	11	21	2000m4–2023m6	95.7
SG	980	9	29	2001m11–2023m6	99.2
PL	744	12	18	2001m10–2023m6	73.9

NOTE. Coverage of the bond-month sample on which the credit-spread pricing regression is estimated, sorted by the number of observations. "Obs." is the number of bond-months; "Firms" and "Bonds" are the numbers of distinct issuers and distinct bonds observed at any point in the sample; "Sample" gives the first and last months observed; "Cov." is the number of months observed as a percentage of the span between them. Counts are computed on the estimation sample, that is, after the filters of this subsection and after dropping bond-months whose default probability is zero and therefore has no log.

B.2 Construction of the country EBP panel

The pricing regression delivers a bond-level excess bond premium for every bond-month in the estimation sample, and a country EBP is the unweighted mean of these across the bonds outstanding in a given country and month. Whether the resulting series is informative as a time series is a separate question from whether its bonds were usable in the pricing regression, and we impose two further requirements before a country enters the analysis of Section 3. These conditions apply to the country EBP panel only: they do not affect the pricing regression, and they remove no country from the database, which we release in full alongside the analysis panel.

False starts. A number of countries have a short early block of observations separated from the main body of the series by a gap of many years, an artefact of a handful of bonds

being outstanding before the local market develops. Argentina is the clearest case: twenty-three months in 1998–99, and then nothing until December 2011. Treating such a block as part of the sample makes the series appear badly incomplete when the modern run is in fact uninterrupted. We therefore treat a leading block separated from the remainder by a gap of more than 36 months as a false start, and date the country's effective sample from the restart.

Length and completeness. On the effective sample so defined, we require at least 120 monthly observations and a within-span coverage rate of at least 80 percent. The rule is stated on these two margins only. A series that is short but uninterrupted is informative about the common component; one riddled with long interior holes is not, however long its nominal span.

Table B.2 reports the resulting panel. Of the 41 countries for which an EBP can be computed, 41 satisfy these conditions and enter the analysis; 4 have a leading false-start block trimmed.

Table B.2 Coverage of the Country EBP Panel

Country	Obs.	Sample	Cov.	Gap	False start	In
AU	306	1998m1–2023m6	100.0	0	–	Yes
CA	306	1998m1–2023m6	100.0	0	–	Yes
CH	306	1998m1–2023m6	100.0	0	–	Yes
CL	306	1998m1–2023m6	100.0	0	–	Yes
DE	306	1998m1–2023m6	100.0	0	–	Yes
ES	306	1998m1–2023m6	100.0	0	–	Yes
FR	306	1998m1–2023m6	100.0	0	–	Yes
GB	306	1998m1–2023m6	100.0	0	–	Yes
HK	306	1998m1–2023m6	100.0	0	–	Yes
JP	306	1998m1–2023m6	100.0	0	–	Yes
NL	306	1998m1–2023m6	100.0	0	–	Yes
NO	306	1998m1–2023m6	100.0	0	–	Yes
SE	306	1998m1–2023m6	100.0	0	–	Yes
US	306	1998m1–2023m6	100.0	0	–	Yes
AT	295	1998m12–2023m6	100.0	0	–	Yes
KR	292	1999m3–2023m6	100.0	0	–	Yes
MY	291	1998m1–2023m6	95.1	15	–	Yes
PT	286	1999m4–2023m6	98.3	5	–	Yes
FI	285	1999m10–2023m6	100.0	0	–	Yes
IT	277	2000m6–2023m6	100.0	0	–	Yes
MX	275	1998m1–2023m6	89.9	31	–	Yes
NZ	267	2000m4–2023m6	95.7	9	–	Yes
GR	263	2000m2–2023m6	93.6	18	–	Yes

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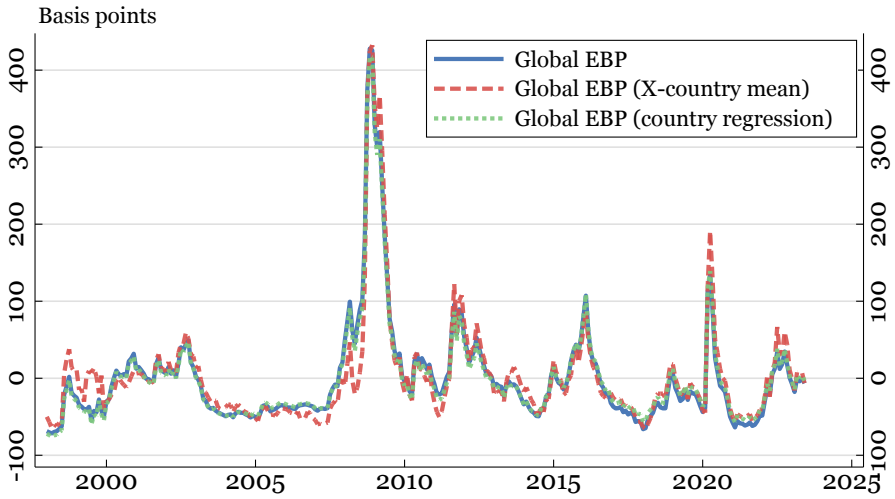
Country	Obs.	Sample	Cov.	Gap	False start	In
BE	261	2001m4–2023m6	97.8	6	–	Yes
SG	258	2001m11–2023m6	99.2	2	–	Yes
LU	241	2003m6–2023m6	100.0	0	Yes	Yes
BR	236	2001m5–2023m6	88.7	30	–	Yes
ZA	232	2002m6–2023m6	91.7	21	–	Yes
TH	227	2004m8–2023m6	100.0	0	–	Yes
IN	197	2007m2–2023m6	100.0	0	–	Yes
RU	189	2006m7–2022m3	100.0	0	–	Yes
CN	186	2008m1–2023m6	100.0	0	–	Yes
IE	186	2008m1–2023m6	100.0	0	Yes	Yes
DK	170	2009m5–2023m6	100.0	0	–	Yes
AE	168	2009m6–2023m6	99.4	1	–	Yes
PL	151	2010m12–2023m6	100.0	0	Yes	Yes
CO	145	2011m6–2023m6	100.0	0	–	Yes
AR	139	2011m12–2023m6	100.0	0	Yes	Yes
ID	139	2010m11–2023m6	91.4	13	–	Yes
PE	139	2011m12–2023m6	100.0	0	–	Yes
TR	128	2012m11–2023m6	100.0	0	–	Yes

NOTE. Coverage of the monthly country EBP panel, sorted by the number of observations. "Obs." is the number of months for which the country EBP is observed on the effective sample; "Sample" gives the effective first and last months; "Cov." is the number of observations as a percentage of the span between them; "Gap" is the longest run of consecutive missing months strictly inside that span. "False start" indicates a country whose effective sample is dated from a restart because a leading block, separated from the main body of the series by a gap of more than 36 months, was treated as a false start; all figures shown are for the resulting effective sample. "In" indicates whether the country enters the analysis sample, which requires at least 120 observations and coverage of at least 80 percent.

Appendix C: GEBP – Additional Results

This appendix collects additional material on the construction of the global EBP. Figure C.1 compares the baseline global series with the two alternative constructions discussed in Section 3.1. Table C.1 reports the full set of country-by-country pricing regressions, summarised in column (2) of Table 2.

Figure C.1 Alternative Constructions of the Global EBP



Note. Three constructions of the global EBP. Monthly, January 1998 to June 2023, in basis points. "Global EBP" (solid) is the baseline of equation (4): bond-level premia from the pooled pricing regression (1), averaged across all outstanding bonds. "Global EBP (country regression)" (short-dashed) replaces the pooled regression with country-by-country estimation, in which every slope is allowed to differ across countries, and averages the resulting bond-level premia across all outstanding bonds. "Global EBP (X-country mean)" (dashed) averages the country EBPs obtained from those same country-by-country regressions, so that every country receives equal weight.

Table C.1 Credit-Spread Pricing Regressions Estimated Country by Country

Country	PD	Age	Face value	Duration	Coupon	Obs.	R^2	Firms	Bonds
AE	0.17***	0.04	-0.17	0.25***	-0.02	1,230	0.85	9	25
	(0.04)	(0.03)	(0.15)	(0.03)	(0.09)				
AR	0.08**	0.07***	-0.11***	0.14***	0.22	1,609	0.64	20	43
	(0.04)	(0.03)	(0.03)	(0.03)	(0.16)				
AT	0.23***	-0.00	0.11	0.24***	0.12**	2,739	0.73	13	48
	(0.05)	(0.02)	(0.10)	(0.03)	(0.06)				
AU	0.06***	0.02	0.01	0.32***	0.18***	18,897	0.61	107	418
	(0.01)	(0.01)	(0.03)	(0.02)	(0.05)				

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Country	PD	Age	Face value	Duration	Coupon	Obs.	R^2	Firms	Bonds
BE	0.11***	0.00	0.01	0.37***	0.26***	8,564	0.67	23	166
	(0.01)	(0.01)	(0.03)	(0.05)	(0.03)				
BR	0.22***	-0.02	-0.00	0.24***	0.49***	12,624	0.59	59	236
	(0.04)	(0.02)	(0.03)	(0.03)	(0.10)				
CA	0.04***	0.06***	0.07***	0.40***	0.11**	89,342	0.61	303	1,746
	(0.01)	(0.01)	(0.02)	(0.02)	(0.05)				
CH	0.08***	0.03**	-0.01	0.36***	0.08***	16,192	0.67	46	440
	(0.03)	(0.01)	(0.06)	(0.02)	(0.02)				
CL	0.12***	-0.04	-0.21***	0.29***	0.38***	3,441	0.54	31	62
	(0.03)	(0.03)	(0.07)	(0.07)	(0.10)				
CN	0.10***	0.00	-0.05*	0.22***	0.41***	17,056	0.86	148	556
	(0.01)	(0.01)	(0.03)	(0.02)	(0.05)				
CO	0.14***	0.01	0.07	0.30***	0.46***	1,483	0.67	13	32
	(0.03)	(0.02)	(0.05)	(0.08)	(0.10)				
DE	0.10***	0.00	0.03	0.29***	0.07**	64,805	0.58	116	1,606
	(0.02)	(0.01)	(0.02)	(0.02)	(0.03)				
DK	0.12***	-0.01	-0.13***	0.35***	0.08***	2,525	0.59	11	62
	(0.03)	(0.01)	(0.03)	(0.05)	(0.03)				
ES	0.21***	-0.04***	0.14***	0.28***	0.21***	18,071	0.60	38	306
	(0.03)	(0.01)	(0.03)	(0.02)	(0.03)				
FI	0.18***	0.09***	0.12*	0.35***	-0.06	4,438	0.56	20	84
	(0.04)	(0.03)	(0.07)	(0.05)	(0.06)				
FR	0.16***	0.00	0.02	0.30***	0.03	67,018	0.62	116	1,226
	(0.02)	(0.01)	(0.03)	(0.02)	(0.02)				
GB	0.10***	0.01	0.00	0.33***	0.12***	85,412	0.63	250	1,652
	(0.01)	(0.01)	(0.02)	(0.02)	(0.03)				
GR	0.19***	0.08	0.08	0.31***	0.46***	1,366	0.86	11	34
	(0.02)	(0.07)	(0.14)	(0.03)	(0.05)				
HK	0.05***	-0.01	-0.03	0.18***	0.26***	6,475	0.70	71	198

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Country	PD	Age	Face value	Duration	Coupon	Obs.	R^2	Firms	Bonds
	(0.01)	(0.03)	(0.04)	(0.05)	(0.07)				
ID	0.01	0.04	-0.16**	0.23***	0.33	1,119	0.71	18	37
	(0.03)	(0.03)	(0.07)	(0.08)	(0.27)				
IE	0.12***	0.00	-0.10**	0.33***	0.09	2,834	0.72	18	77
	(0.02)	(0.03)	(0.05)	(0.05)	(0.08)				
IN	0.07*	-0.05**	-0.12**	0.20***	0.37***	4,959	0.58	26	89
	(0.04)	(0.02)	(0.06)	(0.06)	(0.11)				
IT	0.20***	0.00	0.05**	0.32***	0.07***	21,015	0.60	40	354
	(0.03)	(0.02)	(0.02)	(0.02)	(0.03)				
JP	0.14***	0.02	-0.08*	0.15***	-0.01	119,565	0.64	138	2,229
	(0.01)	(0.01)	(0.04)	(0.02)	(0.03)				
KR	0.14***	0.01	-0.05	0.16***	0.22***	9,565	0.53	37	246
	(0.02)	(0.01)	(0.05)	(0.03)	(0.04)				
LU	0.17***	0.08**	-0.17***	0.32***	0.00	1,545	0.76	28	67
	(0.03)	(0.03)	(0.06)	(0.08)	(0.12)				
MX	0.05***	0.02	-0.07***	0.33***	0.16***	10,199	0.79	59	199
	(0.02)	(0.01)	(0.03)	(0.02)	(0.06)				
MY	0.09**	0.07	0.03	0.09***	0.09	1,607	0.53	11	22
	(0.04)	(0.06)	(0.05)	(0.03)	(0.20)				
NL	0.11***	0.04***	0.06**	0.41***	0.05	18,336	0.71	81	417
	(0.01)	(0.02)	(0.03)	(0.05)	(0.04)				
NO	0.16***	0.06**	-0.04	0.32***	0.01	3,566	0.81	14	77
	(0.05)	(0.03)	(0.04)	(0.04)	(0.04)				
NZ	0.10***	0.10***	-0.61***	0.34***	0.23***	984	0.63	11	21
	(0.02)	(0.02)	(0.07)	(0.03)	(0.07)				
PE	0.10***	-0.02	-0.02	0.37***	0.54***	1,307	0.71	12	22
	(0.02)	(0.04)	(0.08)	(0.07)	(0.08)				
PL	0.12**	0.03	-0.38***	0.44***	0.03	744	0.80	12	18
	(0.05)	(0.02)	(0.09)	(0.07)	(0.20)				

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Country	PD	Age	Face value	Duration	Coupon	Obs.	R^2	Firms	Bonds
PT	0.17***	-0.06*	0.16*	0.29***	0.42***	2,733	0.48	9	49
	(0.04)	(0.03)	(0.08)	(0.06)	(0.14)				
RU	0.19***	-0.04**	-0.05**	0.22***	0.25***	7,613	0.46	18	126
	(0.06)	(0.01)	(0.02)	(0.03)	(0.08)				
SE	0.16***	0.01	0.10*	0.29***	-0.05	6,312	0.56	36	156
	(0.04)	(0.03)	(0.06)	(0.03)	(0.04)				
SG	0.06***	0.05	-0.01	0.20***	0.08	980	0.82	10	30
	(0.01)	(0.04)	(0.14)	(0.07)	(0.05)				
TH	0.14***	0.01	-0.02	0.19***	0.45***	1,830	0.70	11	32
	(0.04)	(0.04)	(0.09)	(0.05)	(0.12)				
TR	0.10***	0.01	-0.41***	0.28***	0.06	1,351	0.61	10	22
	(0.02)	(0.02)	(0.14)	(0.04)	(0.10)				
US	0.06***	0.01*	-0.04***	0.34***	0.35***	825,388	0.72	2,304	19,774
	(0.00)	(0.00)	(0.01)	(0.01)	(0.03)				
ZA	0.15***	-0.00	0.09	0.25***	0.24**	2,566	0.62	12	59
	(0.03)	(0.03)	(0.11)	(0.06)	(0.11)				

NOTE. The table reports the bond-level credit-spread pricing regressions used to construct the EBP, estimated separately for each country. The dependent variable is the log option-adjusted spread; regressors are the log 12-month default probability (PD), log age, log USD face value, log effective duration, and log coupon, with currency, rating, and industry (ind4) fixed effects. Standard errors, two-way clustered by firm and month, are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Coefficients corresponding to the constant and fixed effects are not reported.

Appendix D: Common Factors

D.1 The EM principal-components estimator

Let x_{it} denote the EBP of country i in month t, in basis points, observed on a set $\mathcal{S}$ of country-months that does not cover the full $N \times T$ rectangle. Each series is first standardized over the months on which it is observed,

$$z_{it} = \frac{x_{it} - \mu_i}{\sigma_i}, \qquad \mu_i = \frac{1}{T_i} \sum_{t \in \mathcal{S}_i} x_{it}, \qquad \sigma_i^2 = \frac{1}{T_i} \sum_{t \in \mathcal{S}_i} (x_{it} - \mu_i)^2, \tag{D.1}$$

so that the decomposition is correlation-based and every country enters with equal weight, whatever the volatility of its EBP. The approximate factor model is $z_{it} = \lambda_i' F_t + e_{it}$, with r factors.

Were the panel balanced, $\widehat{\Lambda}$ would be the matrix of eigenvectors associated with the r largest eigenvalues of the second-moment matrix of Z, normalized so that $\widehat{\Lambda}'\widehat{\Lambda} = I_r$, and $\widehat{F} = Z\widehat{\Lambda}$. It is not, so we use the EM algorithm of Stock and Watson (2002), which alternates between extracting principal components from a completed panel and re-imputing the missing cells from the resulting common component. Writing $Z^{(0)}$ for the standardized panel with the missing cells set to zero, that is, to the country mean, iteration k computes $\widehat{\Lambda}^{(k)}$ and $\widehat{F}^{(k)} = Z^{(k-1)}\widehat{\Lambda}^{(k)}$ from $Z^{(k-1)}$ exactly as in the balanced case, forms the common component $C^{(k)} = \widehat{F}^{(k)}\widehat{\Lambda}^{(k)\prime}$, and sets

$$z_{it}^{(k)} = \begin{cases} z_{it} & \text{if } (i,t) \in \mathcal{S}, \\ c_{it}^{(k)} & \text{otherwise.} \end{cases} \tag{D.2}$$

Observed cells are never overwritten; only the missing ones move. The iteration stops when the largest absolute change in any imputed cell falls below 10^{-7}, and the factor is sign-normalized so that it is positively related to the cross-sectional mean of the observed standardized data. The procedure handles ragged edges and interior gaps alike, which matters here because the two forms of missingness are not equally common: countries predominantly enter the panel late rather than dropping in and out of it.

The statistic we report is the share of the variance of the *observed* cells that the common component explains,

$$R^2_{\mathcal{S}} = 1 - \frac{\sum_{(i,t)\in\mathcal{S}} (z_{it} - c_{it})^2}{\sum_{(i,t)\in\mathcal{S}} z_{it}^2}, \tag{D.3}$$

and not the eigenvalue shares of the completed panel. The distinction is not cosmetic: the imputed cells are by construction exactly rank r, so any statistic computed on the completed panel is mechanically inflated by the imputation, and the more so the more of the panel is missing. The same expression restricted to a single column i gives the country variance shares plotted in Panel B of Figure 4.

Finally, the estimated factor is rescaled to basis points by matching the first two moments

of GEBP, $F_t^{bp} = m_g + (\widehat{F}_t - m_f) s_g / s_f$, where (m_f, s_f) and (m_g, s_g) are the mean and standard deviation of $\widehat{F}$ and of GEBP. Country i's loading in basis points of local EBP per basis point of the global factor, the object plotted in Panel A of Figure 4, is then $b_i = \sigma_i \lambda_i s_f / s_g$.

D.2 Loadings from country-by-country regressions

Figure 4 in Section 3.2 reads each country's loading on the global common component, and the share of its EBP variance that the component explains, directly off the EM principal-components decomposition reported in column 4 of Table 3. Both objects are therefore internal to a joint decomposition of the panel. An alternative and more familiar characterisation projects each country EBP on the estimated factor one series at a time,

$$EBP_{it} = a_i + b_i \, F_t + u_{it}, \tag{D.4}$$

where F_t is the estimated global factor expressed in basis points, and reads amplification off the slope b_i and synchronization off the regression R^2. Figure D.1 reports the two.

The two constructions differ only in that the first imposes the single-factor structure on the whole panel at once, while the second treats the estimated factor as an observed regressor and fits each country separately. In practice they are all but interchangeable: across countries the regression slopes correlate with the loadings at 0.999 and the regression R^2 values with the variance shares at 0.998. Nothing in Section 3.2 depends on which of the two is used.

D.3 The completed database

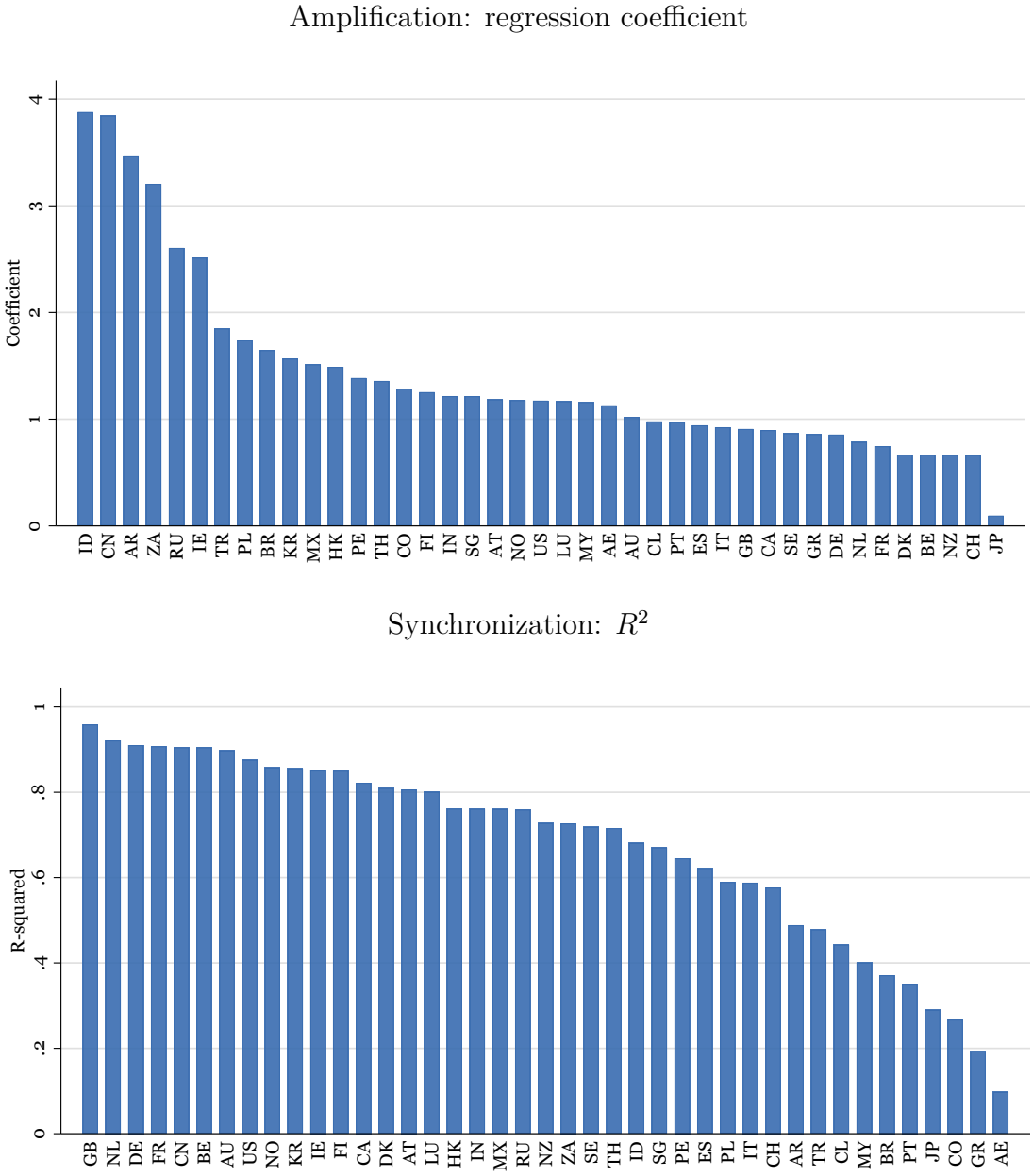
The country EBP panel is unbalanced, while many uses of it require a rectangular one. We therefore release, alongside the analysis panel of Appendix B.2 and the complete panel that precedes it, a *completed* panel in which every country of the analysis sample is observed in every month of the sample window. The missing cells are filled with the fitted common component of the one-factor decomposition described above, that is, with the completed panel (D.2) that the EM algorithm converges to, expressed back in basis points:

$$\widetilde{EBP}_{it} = \begin{cases} EBP_{it} & \text{if } (i,t) \in \mathcal{S}, \\ \mu_i + \sigma_i \lambda_i \widehat{F}_t \;=\; a_i + b_i F_t^{bp} & \text{otherwise,} \end{cases} \tag{D.5}$$

with b_i the loading of Figure 4. The filled series and the loadings reported in Section 3.2 are thus read off one and the same estimate, and the fill requires no assumption that the factor analysis has not already made. Of the 12,546 country-months in the analysis panel, 2,339 are filled in this way, or 18.6 percent; 27 of the 41 countries are affected, and at most 58.2 percent of any single country's series is fitted rather than observed. The released file carries a flag of identical shape marking every filled cell.

Three properties of the fill bound what it can be used for, and we state them plainly rather than leaving them to be discovered.

Figure D.1 Sensitivity of Country EBPs to the Global Factor: Regression Estimates



Note. Slope coefficient (upper panel) and R^2 (lower panel) from country-by-country time-series regressions of the local EBP on the estimated global factor and a constant, both expressed in basis points. Estimated over each country's observed months, January 1998 to June 2023, for the 41 countries of the analysis panel. Countries are ordered from the largest to the smallest value in each panel. The factor-decomposition counterpart of this figure is Figure 4.

First, it is a pure common component. The idiosyncratic part e_{it} is not modelled, so a filled segment is a scaled copy of the global factor and understates the volatility of the underlying series by exactly the share of variance the factor leaves unexplained – a share that, by Panel B of Figure 4, is large for precisely the countries with the most to fill. For a weakly loading country such as Japan a filled segment would be close to a straight line.

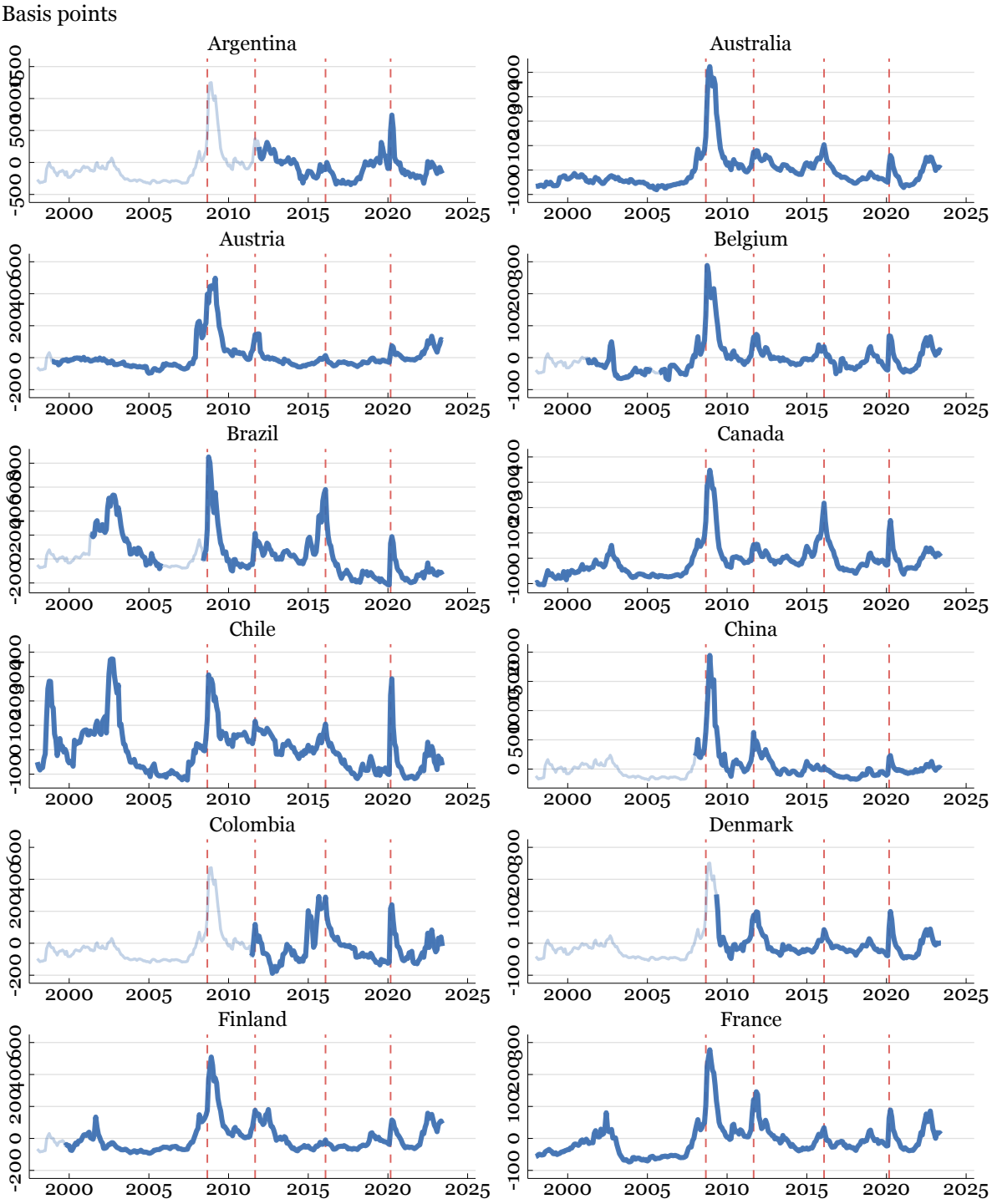
Second, the fitted values are not bounded by the range of the data actually observed for that country, and for 9 countries they are not contained in it. This is a property of the exercise rather than a defect of it: a country that enters the panel after 2008 has its pre-entry months reconstructed from a factor that peaked in a period over which the country itself is unobserved, and the reconstruction is larger than anything the country was later seen to record. It does, however, mean that the filled segments are extrapolations in a stronger sense than the interior gaps are.

Third, the leading false-start blocks trimmed in Appendix B.2 are not restored. Those months are treated as missing and imputed like any other, which is the point of the trim: the observations exist, but the series they belong to is not the one the country's modern bond market generates. Users who prefer the raw observations have them in the complete panel, which is released untouched.

The fill is, finally, not sensitive to whether the single-factor structure is imposed on the panel jointly or series by series. Replacing (D.5) with the fitted values of the country-by-country regressions of Appendix D.2 changes the imputed cells by little: the two sets of fitted values correlate at 0.987. We recommend the analysis panel for estimation in any case, and the completed panel only where a rectangular one is unavoidable.

Figure D.2 shows the result country by country, and is also the most direct way to see what the database contains: the length of each observed series, where the gaps fall, and what the fill puts in their place.

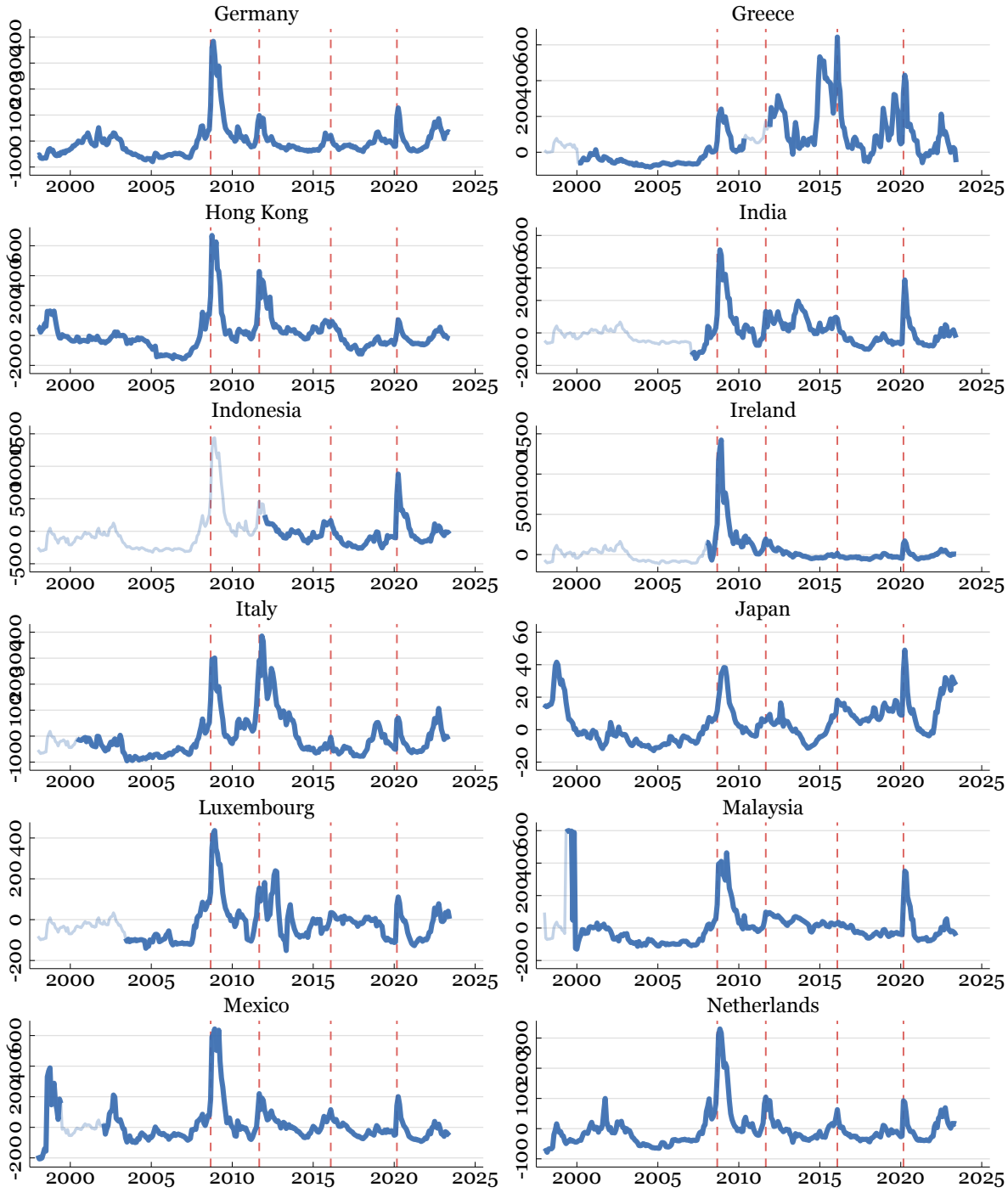
Figure D.2 Country EBPs, Observed and Filled with the Global Factor



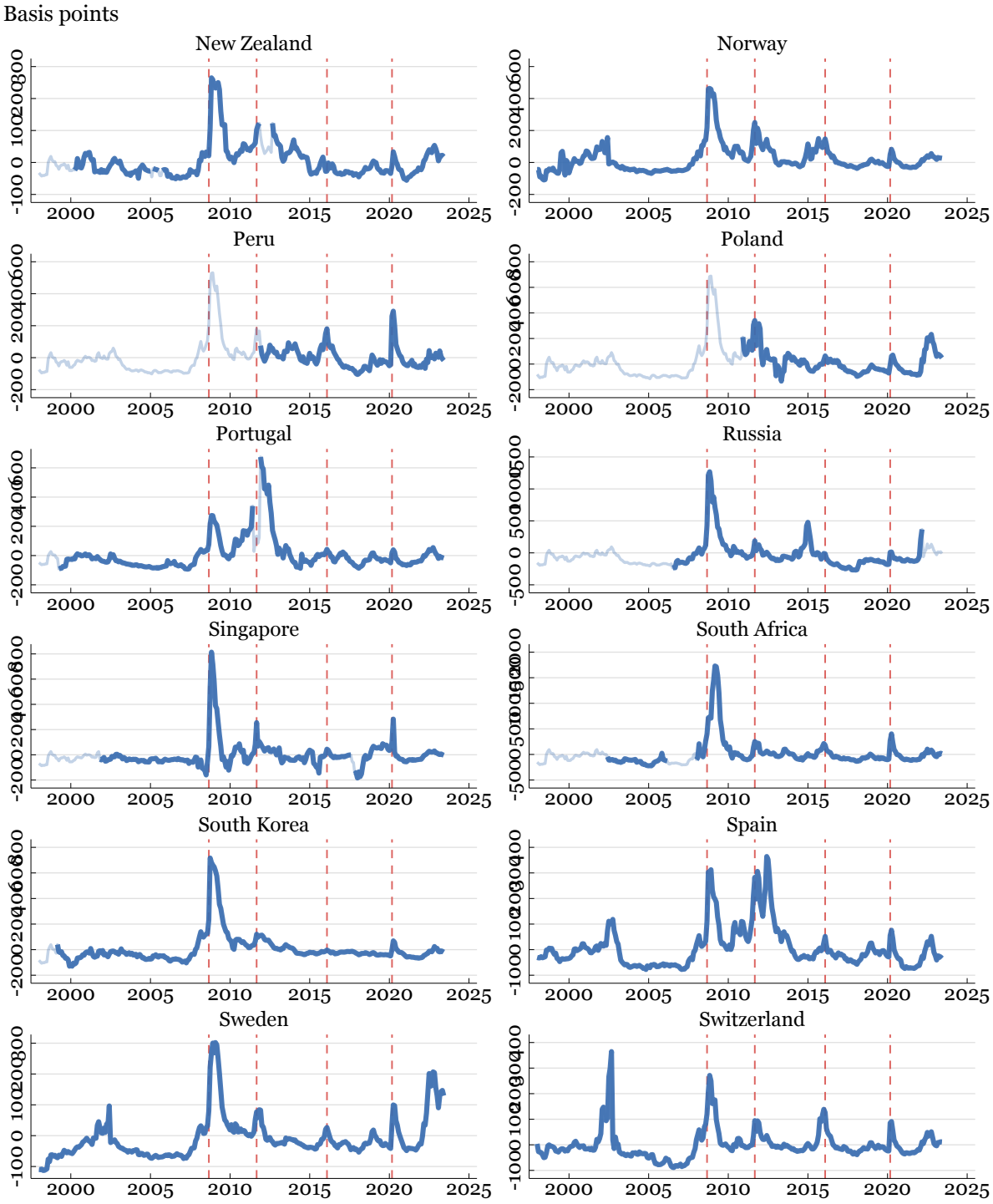
Note. Monthly EBP of each of the 41 countries of the analysis panel, in basis points, January 1998 to June 2023. The dark line is the observed series; the light line is the completed series of equation (D.5), so that light segments are the months supplied by the fill and dark segments the months the country is observed. Vertical lines mark the four episodes dated in Figure 3: the collapse of Lehman Brothers, the escalation of the euro-area sovereign debt crisis, the oil and China episode, and the onset of the COVID-19 pandemic. *The vertical scale is free and differs across panels*: country EBPs span two orders of magnitude, and a common scale would flatten most of the panels to a line. Levels are therefore not comparable across panels by eye; Figure 3 plots all countries on one scale. Countries are ordered alphabetically by name and the figure continues over the following pages.

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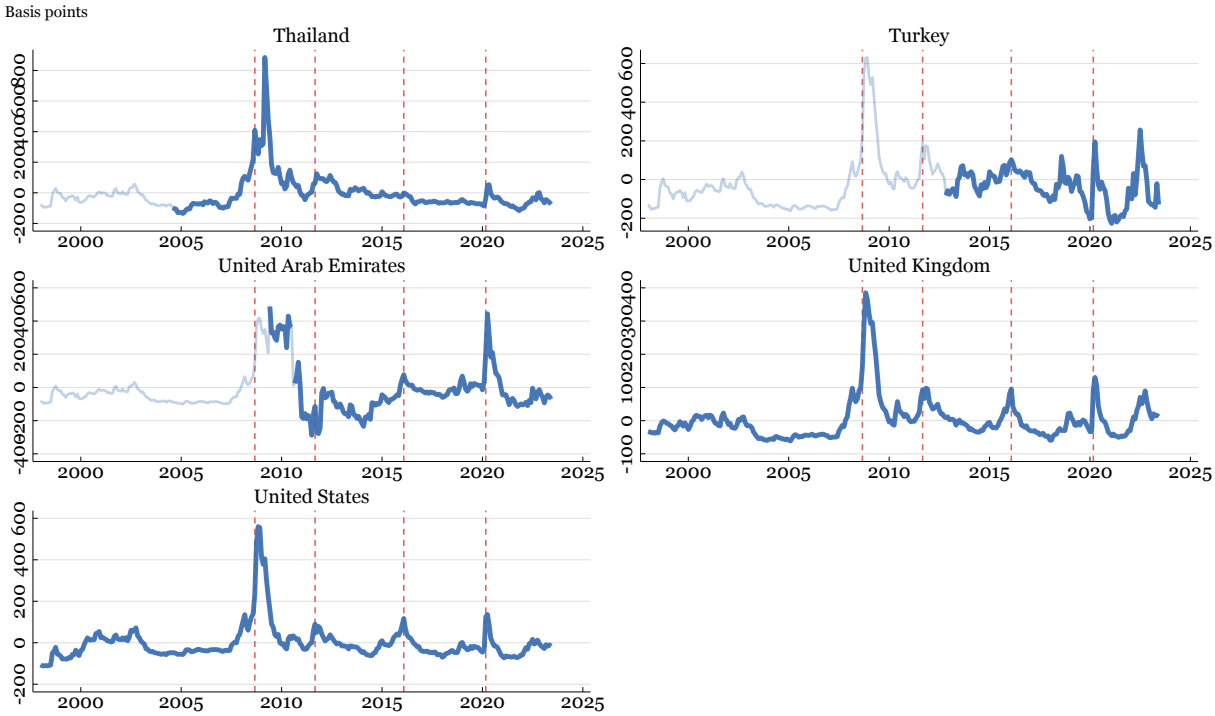
Basis points



(CONTINUED)


Basis points
New Zealand
Norway
Peru
Poland
Portugal
Russia
Singapore
South Africa
South Korea
Spain
Sweden
Switzerland
2000
2005
2010
2015
2020
2025

(CONTINUED)


Basis points
Thailand
-200 0 200 400 600 800
Turkey
-200 0 200 400 600
United Arab Emirates
-400 -200 0 200 400 600
United Kingdom
-100 0 100 200 300 400
United States
-200 0 200 400 600
2000
2005
2010
2015
2020
2025

Appendix E: Validation of the EBP series

Compared to the original US-EBP series constructed by Gilchrist and Zakrajšek (2012), ours differs by taking default probabilities (PDs) from the NUS-CRI database (rather than obtaining PDs from the Merton (1974)-based route). The NUS-CRI database calculates PDs from a multivariate model (Duan et al. (2012)) that goes beyond a sole reliance on Merton's DD (which ends up as being just one of many inputs). While this is theoretically appealing (since prior research has shown that DD variation by itself is insufficient to explain observed PD variation; see Bharath and Shumway (2008), Campbell et al. (2008), and Duffie et al. (2007))), it does beg the question how the resulting EBPs compare – both when it comes to their behavior over time, as well as with respect to their relation to real activity. This appendix addresses that question, first for the US (Section E.1) and then for the UK (Section E.2), the two countries for which an externally constructed benchmark series is available. In both cases, our validation exercises provide comfort that our country-level EBP series are capturing the relevant dimensions of corporate bond market sentiment.

E.1 Validation on US data

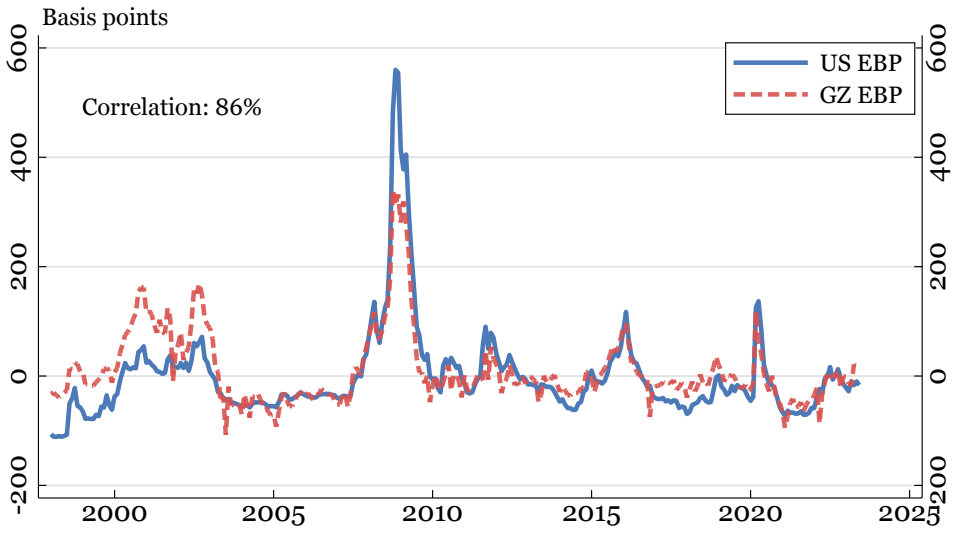
The US provides the most natural ground to validate our approach, since the US-EBP series is well tested and established. To this end, Figure E.1 plots the original US-EBP series ('US-EBP(GZ)', taken from Gilchrist and Zakrajšek (2012)) alongside its equivalent that results from our NUS-CRI-based approach. As can be seen from this figure, the two series show a high degree of co-movement (with a correlation of 0.86 in levels and 0.63 in first differences) which provides a first form of validation of our US-EBP series.

As a second validation step, we turn to an important feature of the US-EBP series emphasized by Gilchrist and Zakrajšek (2012): its ability to forecast real activity. This feature has led to the EBP's status as being seen as a useful indicator of financial conditions – used accordingly by central banks worldwide and routinely included as a helpful control variable in dynamic analyses of monetary policy shocks (see, e.g., Gertler and Karadi (2015)).

In this vein, we compare the forecasting performance of our US-EBP to the US-EBP(GZ). We do this for the original sample underpinning Gilchrist and Zakrajšek (2012), spanning January 1997 to September 2010 (but results are robust to using an extended sample). Following Gilchrist and Zakrajšek (2012) we consider a forecasting regression which includes the US-EBP measure alongside the fitted component of the spread ($\hat{S}$) as well as various proxies for the monetary policy stance, namely the term spread (TS, defined as the difference between the 10-year and 3-month sovereign borrowing rates), and the real policy rate (RFF). As outcome variables (Y), we look at the natural log of private nonfarm payroll employment (EMP), the level of the unemployment rate (UER), and the natural log of industrial production in the manufacturing sector (IPM). We thus estimate:

$$\nabla^h Y_{t+h} = \sum_{i=1}^{p} \beta_i \Delta Y_{t-i} + \gamma_1 TS_t + \gamma_2 RFF_t + \gamma_3 \hat{S}_t + \gamma_4 EBP_t + \varepsilon_{t+h}, \tag{E.1}$$

Figure E.1 US EBP: NUS-CRI-Based Series versus the Federal Reserve Series



Note. The US EBP constructed here (solid) against the excess bond premium of Gilchrist and Zakrajšek (2012) as maintained by the Federal Reserve Board (dashed), both in basis points and both plotted over the sample the two series have in common. The two constructions share the same decomposition of the corporate spread but differ in the measure of default risk: ours uses the NUS-CRI 12-month default probability, theirs the Merton (1974) distance to default. The annotation reports the correlation between the two series in levels; the correlation in first differences is 0.63. Sources: LSEG/BAML bond data, NUS-CRI default probabilities, and the Federal Reserve Board EBP file.

with $\nabla^h Y_{t+h} := \frac{d}{h+1}(Y_{t+h} - Y_{t-1})$, $h > 0$ denoting the forecast horizon, and $d = 1200$ representing a scaling constant (reflecting the monthly frequency of the data).

Table E.1 reports the results when using our new US-EBP measure, while results that follow from usage of US-EBP(GZ), the original series by Gilchrist and Zakrajšek (2012), are reported in Table E.2. Both US-EBP measures exhibit significant explanatory power for future economic development across variables and horizons, and no further information is contained in the fitted component of the spread.

Interestingly, while the term spread and real federal funds rate remain significant predictors when using the original US-EBP(GZ) series, our US-EBP series fully encompasses the information content of alternative variables capturing the monetary policy stance. Moreover, across models, the coefficients on our new US-EBP measure surpass the coefficients that result when using US-EBP(GZ). Together, these findings suggest that, if anything, our newly constructed US-EBP series provides an even more complete picture of financial conditions than the original series constructed by Gilchrist and Zakrajšek (2012) . This may reflect the fact that our PD (and hence EBP) measure is arguably more comprehensive since it is not solely based on the Merton-implied DD, but takes account of other covariates as well (which have been shown to matter; recall the discussion in Section 2).

Table E.1 US FORECASTING EXERCISE - USING OUR US-EBP, OVERLAPPING SAMPLE

	(1) EMP_3m	(2) UER_3m	(3) IPM_3m	(4) EMP_12m	(5) UER_12m	(6) IPM_12m
Term spread	0.06	0.01	-0.14	-0.15	0.31	-0.37
	(0.09)	(0.14)	(0.14)	(0.20)	(0.24)	(0.28)
Real FFR	-0.08	0.12	0.08	0.04	-0.09	0.39
	(0.13)	(0.20)	(0.23)	(0.32)	(0.37)	(0.48)
Predicted Spread US	0.03	0.03	-0.06	-0.09	0.27	-0.29
	(0.07)	(0.12)	(0.16)	(0.24)	(0.29)	(0.36)
EBP US	-0.39***	0.61***	-0.47***	-0.40***	0.50***	-0.51**
	(0.04)	(0.06)	(0.09)	(0.12)	(0.12)	(0.23)
R-squared	0.866	0.709	0.659	0.602	0.521	0.382
Obs.	150	150	150	141	141	141

NOTE. Forecasting regressions of the form given in the equation above, using the US EBP constructed here. The dependent variable is the cumulative change in private nonfarm payroll employment (EMP), the unemployment rate (UER), or industrial production in manufacturing (IPM) over the horizon h given in the column heading. Each regression includes the term spread, the real federal funds rate, the fitted component of the corporate spread, the EBP, and lags of the outcome. Estimated on monthly data from January 1998 to September 2010, the sample of Gilchrist and Zakrajšek (2012); results are robust to extending it. All predictors are standardized to have mean zero and unit variance, so that the coefficients are directly comparable across columns and with Table E.2.

Table E.2 US FORECASTING EXERCISE - USING US-EBP(GZ), OVERLAPPING SAMPLE

	(1) EMP_3m	(2) UER_3m	(3) IPM_3m	(4) EMP_12m	(5) UER_12m	(6) IPM_12m
Term spread	-0.16**	0.34***	-0.35***	-0.36***	0.53***	-0.54***
	(0.06)	(0.11)	(0.10)	(0.12)	(0.15)	(0.19)
Real FFR	0.28**	-0.49***	0.45***	0.45**	-0.60**	0.63*
	(0.11)	(0.17)	(0.17)	(0.20)	(0.25)	(0.36)
Predicted Spread US	-0.01	-0.08	-0.11	-0.07	-0.12	-0.31*
	(0.08)	(0.14)	(0.12)	(0.10)	(0.12)	(0.16)
EBP US	-0.25***	0.37***	-0.33***	-0.41***	0.46***	-0.50***
	(0.05)	(0.07)	(0.08)	(0.07)	(0.08)	(0.16)
R-squared	0.855	0.666	0.656	0.685	0.558	0.431
Obs.	150	150	150	141	141	141

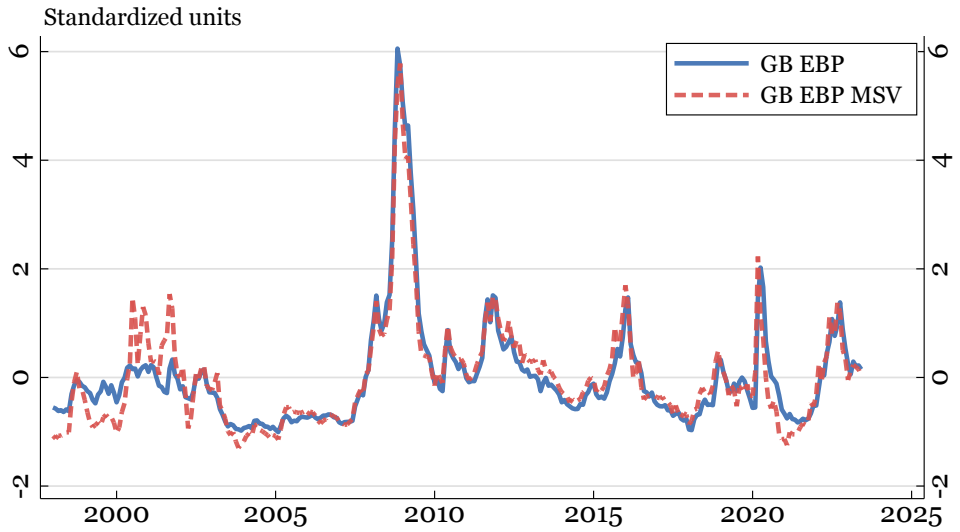
NOTE. The counterpart of Table E.1, replacing the US EBP constructed here with the original series of Gilchrist and Zakrajšek (2012). Specification, sample and standardization are otherwise identical: the dependent variable is the cumulative change in private nonfarm payroll employment (EMP), the unemployment rate (UER), or industrial production in manufacturing (IPM) over the horizon h given in the column heading; the regressors are the term spread, the real federal funds rate, the fitted component of the corporate spread, the EBP, and lags of the outcome; the sample runs from January 1998 to September 2010 and all predictors are standardized to have mean zero and unit variance.

E.2 Validation on UK data

The exercises documented in the previous subsection comfort us that the US-EBP series that follows from deploying NUS-CRI PD data yields sensible results for the US. In this subsection, we make the case that our non-US EBP series are similarly informative on corporate bond market sentiment in the other countries in our sample. The UK is the only non-US country for which an independently constructed EBP series is available, and it therefore provides the sole direct out-of-sample check on our approach.

Maqui et al. (2025) follow Gilchrist and Zakrajšek (2012) in using the Merton (1974)-based approach to PD/EBP construction for the UK. As can be seen in Figure E.2, their resulting UK-EBP series shows a tight comovement with ours (with a correlation of 0.95 in levels and 0.76 in first differences).

Figure E.2 UK EBP: NUS-CRI-Based Series versus Maqui et al. (2025)



Note. The UK EBP constructed here (solid) against the UK excess bond premium of Maqui et al. (2025) (dashed), plotted over the sample the two series have in common. Both series are standardized to have mean zero and unit variance over that sample, since the two constructions are not on a common scale. The two differ in the measure of default risk: ours uses the NUS-CRI 12-month default probability, theirs the Merton (1974) distance to default. The correlation between the two series is 0.95 in levels and 0.76 in first differences. Sources: LSEG/BAML bond data, NUS-CRI default probabilities, and the series of Maqui et al. (2025).

The forecasting evidence establishing the economic content of the non-US series – namely that country-level EBPs help predict domestic real activity – is presented in the main text (Section 3.3), with additional country-level forecasting tables reported in Appendix F.

Appendix F: Forecasting exercise

Table F.1 Forecasting exercise - 12-month ahead unemployment rate - country and global EBP

	(1) AU	(2) CA	(3) DE	(4) FR	(5) GB	(6) JP	(7) US
Term spread	-0.42***	-0.26	-0.30***	0.24***	-0.26***	0.25***	-0.22
	(0.09)	(0.23)	(0.06)	(0.09)	(0.09)	(0.08)	(0.17)
Real policy rate	0.00	0.45**	-0.09	0.03	-0.00	-0.18*	0.46**
	(0.12)	(0.20)	(0.07)	(0.15)	(0.10)	(0.10)	(0.18)
Predicted Spread	0.39*	-0.41	-0.61**	1.06***	0.29	0.75	0.55*
	(0.20)	(0.25)	(0.25)	(0.31)	(0.29)	(0.50)	(0.29)
EBP_ctry_spec_std	0.08	0.27*	0.05	-0.04	-0.03	0.11***	0.40*
	(0.05)	(0.15)	(0.04)	(0.09)	(0.08)	(0.04)	(0.22)
Global EBP	0.16***	0.54***	0.15***	0.33***	0.23***	0.18***	0.62***
	(0.05)	(0.08)	(0.03)	(0.05)	(0.07)	(0.03)	(0.12)
R-squared	0.413	0.297	0.464	0.580	0.387	0.302	0.353
Obs.	282	282	174	222	249	287	282

Note. Columns are countries; the dependent variable is the change in the unemployment rate over the following 12 months. The excess bond premium enters as two orthogonal terms: the global EBP, and the country-specific EBP, obtained as the residual from a regression of the country EBP on the global EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.2 Forecasting exercise - US - country EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	-0.10	0.07**	-0.23	0.12***
	(0.17)	(0.03)	(0.17)	(0.04)
Real policy rate	0.21*	0.03	0.46**	-0.02
	(0.11)	(0.03)	(0.18)	(0.04)
Predicted Spread	-0.02	-0.04*	0.39*	-0.16***
	(0.07)	(0.02)	(0.23)	(0.05)
Ctry EBP	0.48***	-0.17***	0.51***	-0.17***
	(0.09)	(0.03)	(0.10)	(0.04)
R-squared	0.260	0.448	0.336	0.393
Obs.	291	281	282	270

Note. The excess bond premium enters as the country EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.3 Forecasting exercise - US - country and global EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	-0.10	0.08**	-0.22	0.12***
	(0.17)	(0.03)	(0.17)	(0.04)
Real policy rate	0.22*	0.03	0.46**	-0.01
	(0.12)	(0.03)	(0.18)	(0.04)
Predicted Spread	-0.05	-0.06*	0.55*	-0.22***
	(0.10)	(0.03)	(0.29)	(0.08)
Ctry specific EBP	0.05	-0.08*	0.40*	-0.15
	(0.14)	(0.04)	(0.22)	(0.10)
Global EBP	0.67***	-0.22***	0.62***	-0.20***
	(0.14)	(0.04)	(0.12)	(0.04)
R-squared	0.260	0.454	0.353	0.419
Obs.	291	281	282	270

Note. The excess bond premium enters as two orthogonal terms: the global EBP, and the country-specific EBP, obtained as the residual from a regression of the country EBP on the global EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.4 Forecasting exercise - GB - country EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	-0.13***	-0.00	-0.25***	0.01
	(0.04)	(0.04)	(0.09)	(0.07)
Real policy rate	-0.06	0.02	0.03	0.02
	(0.06)	(0.03)	(0.10)	(0.07)
Predicted Spread	-0.06	-0.17	0.27	-0.30
	(0.15)	(0.16)	(0.29)	(0.19)
Ctry EBP	0.21***	-0.15***	0.24***	-0.19***
	(0.04)	(0.05)	(0.08)	(0.05)
R-squared	0.521	0.693	0.376	0.611
Obs.	258	293	249	282

Note. The excess bond premium enters as the country EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.5 Forecasting exercise - GB - country and global EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	-0.14***	0.02	-0.26***	0.03
	(0.04)	(0.04)	(0.09)	(0.07)
Real policy rate	-0.07	0.07*	-0.00	0.05
	(0.06)	(0.04)	(0.10)	(0.07)
Predicted Spread	-0.05	-0.19	0.29	-0.34*
	(0.15)	(0.16)	(0.29)	(0.18)
Ctry specific EBP	0.03	0.05	-0.03	0.04
	(0.04)	(0.05)	(0.08)	(0.05)
Global EBP	0.19***	-0.15***	0.23***	-0.19***
	(0.03)	(0.04)	(0.07)	(0.04)
R-squared	0.523	0.701	0.387	0.618
Obs.	258	293	249	282

Note. The excess bond premium enters as two orthogonal terms: the global EBP, and the country-specific EBP, obtained as the residual from a regression of the country EBP on the global EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.6 Forecasting exercise - JP - country EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	0.17***	0.03	0.29***	0.00
	(0.06)	(0.12)	(0.10)	(0.20)
Real policy rate	0.07	0.17	-0.18*	0.46**
	(0.06)	(0.12)	(0.10)	(0.20)
Predicted Spread	0.16	1.08	0.46	0.81
	(0.35)	(0.93)	(0.49)	(1.08)
Ctry EBP	1.01***	-1.12	1.08***	-1.33**
	(0.27)	(0.71)	(0.25)	(0.62)
R-squared	0.477	0.562	0.240	0.566
Obs.	291	291	290	282

Note. The excess bond premium enters as the country EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.7 Forecasting exercise - JP - country and global EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	0.13**	0.15	0.25***	0.13
	(0.06)	(0.10)	(0.08)	(0.15)
Real policy rate	0.07	0.20**	-0.18*	0.43**
	(0.05)	(0.09)	(0.10)	(0.18)
Predicted Spread	0.55*	-0.01	0.75	0.06
	(0.33)	(0.67)	(0.50)	(0.93)
Ctry specific EBP	0.10***	0.05	0.11***	-0.02
	(0.04)	(0.08)	(0.04)	(0.09)
Global EBP	0.17***	-0.40***	0.18***	-0.35***
	(0.02)	(0.11)	(0.03)	(0.09)
R-squared	0.546	0.688	0.302	0.623
Obs.	291	291	287	282

Note. The excess bond premium enters as two orthogonal terms: the global EBP, and the country-specific EBP, obtained as the residual from a regression of the country EBP on the global EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.8 Forecasting exercise - FR - country EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	0.10	0.11	0.18**	0.27***
	(0.08)	(0.08)	(0.09)	(0.09)
Real policy rate	0.20**	-0.03	0.22*	-0.02
	(0.08)	(0.08)	(0.12)	(0.09)
Predicted Spread	0.77**	-0.03	1.14***	-0.27
	(0.35)	(0.30)	(0.33)	(0.28)
Ctry EBP	0.29***	-0.29***	0.38***	-0.40***
	(0.07)	(0.08)	(0.08)	(0.06)
R-squared	0.734	0.830	0.546	0.821
Obs.	231	291	222	282

Note. The excess bond premium enters as the country EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.9 Forecasting exercise - FR - country and global EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	0.15*	0.09	0.24***	0.27***
	(0.09)	(0.07)	(0.09)	(0.09)
Real policy rate	0.06	0.11	0.03	-0.00
	(0.12)	(0.09)	(0.15)	(0.08)
Predicted Spread	0.71**	0.08	1.06***	-0.26
	(0.35)	(0.29)	(0.31)	(0.26)
Ctry specific EBP	-0.02	0.05	-0.04	-0.13**
	(0.08)	(0.05)	(0.09)	(0.06)
Global EBP	0.25***	-0.28***	0.33***	-0.30***
	(0.04)	(0.04)	(0.05)	(0.05)
R-squared	0.743	0.837	0.580	0.821
Obs.	231	291	222	282

Note. The excess bond premium enters as two orthogonal terms: the global EBP, and the country-specific EBP, obtained as the residual from a regression of the country EBP on the global EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.10 Forecasting exercise - DE - country EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	-0.10	0.16**	-0.30***	0.49***
	(0.06)	(0.08)	(0.06)	(0.09)
Real policy rate	0.00	0.02	-0.08	0.10
	(0.03)	(0.05)	(0.06)	(0.06)
Predicted Spread	-0.77***	0.21	-0.60**	-0.89**
	(0.26)	(0.25)	(0.25)	(0.37)
Ctry EBP	0.24***	-0.37***	0.17***	-0.22***
	(0.04)	(0.08)	(0.04)	(0.06)
R-squared	0.518	0.647	0.464	0.667
Obs.	183	291	174	282

Note. The excess bond premium enters as the country EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.11 Forecasting exercise - DE - country and global EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	-0.09	0.15**	-0.30***	0.50***
	(0.06)	(0.08)	(0.06)	(0.10)
Real policy rate	-0.04	0.06	-0.09	0.05
	(0.04)	(0.06)	(0.07)	(0.06)
Predicted Spread	-0.92***	0.33	-0.61**	-1.10***
	(0.26)	(0.26)	(0.25)	(0.36)
Ctry specific EBP	0.02	-0.08	0.05	-0.16**
	(0.04)	(0.06)	(0.04)	(0.07)
Global EBP	0.26***	-0.37***	0.15***	-0.13**
	(0.04)	(0.08)	(0.03)	(0.06)
R-squared	0.532	0.649	0.464	0.673
Obs.	183	291	174	282

Note. The excess bond premium enters as two orthogonal terms: the global EBP, and the country-specific EBP, obtained as the residual from a regression of the country EBP on the global EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.12 Forecasting exercise - CA - country EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	-0.13	0.10*	-0.26	0.22***
	(0.18)	(0.05)	(0.24)	(0.07)
Real policy rate	0.37***	-0.07	0.45**	-0.08
	(0.12)	(0.05)	(0.20)	(0.08)
Predicted Spread	-0.60**	0.38***	-0.40*	0.57***
	(0.24)	(0.09)	(0.22)	(0.11)
Ctry EBP	0.62***	-0.30***	0.56***	-0.35***
	(0.09)	(0.04)	(0.09)	(0.05)
R-squared	0.407	0.775	0.297	0.607
Obs.	291	291	282	282

Note. The excess bond premium enters as the country EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.13 Forecasting exercise - CA - country and global EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	-0.12	0.11**	-0.26	0.23***
	(0.18)	(0.05)	(0.23)	(0.06)
Real policy rate	0.37***	-0.05	0.45**	-0.05
	(0.14)	(0.05)	(0.20)	(0.08)
Predicted Spread	-0.61**	0.35***	-0.41	0.53***
	(0.26)	(0.10)	(0.25)	(0.12)
Ctry specific EBP	0.29***	-0.10*	0.27*	-0.11*
	(0.11)	(0.06)	(0.15)	(0.06)
Global EBP	0.60***	-0.30***	0.54***	-0.35***
	(0.08)	(0.03)	(0.08)	(0.05)
R-squared	0.407	0.776	0.297	0.609
Obs.	291	291	282	282

Note. The excess bond premium enters as two orthogonal terms: the global EBP, and the country-specific EBP, obtained as the residual from a regression of the country EBP on the global EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.14 Forecasting exercise - AU - country EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	-0.19***	0.12	-0.42***	0.29**
	(0.07)	(0.09)	(0.10)	(0.12)
Real policy rate	-0.05	0.31**	0.00	0.25**
	(0.08)	(0.14)	(0.12)	(0.12)
Predicted Spread	-0.02	0.30	0.40**	-0.33
	(0.16)	(0.18)	(0.20)	(0.21)
Ctry EBP	0.20***	-0.25***	0.14***	-0.15***
	(0.05)	(0.06)	(0.05)	(0.05)
R-squared	0.487	0.790	0.413	0.672
Obs.	291	238	282	229

Note. The excess bond premium enters as the country EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Table F.15 Forecasting exercise - AU - country and global EBP

	(1) UER_3m	(2) IPM_3m	(3) UER_12m	(4) IPM_12m
Term spread	-0.15**	0.09	-0.42***	0.25**
	(0.07)	(0.08)	(0.09)	(0.11)
Real policy rate	-0.04	0.30**	0.00	0.27**
	(0.08)	(0.13)	(0.12)	(0.11)
Predicted Spread	0.09	0.17	0.39*	-0.52**
	(0.17)	(0.19)	(0.20)	(0.20)
Ctry specific EBP	-0.00	0.01	0.08	0.13**
	(0.08)	(0.05)	(0.05)	(0.06)
Global EBP	0.23***	-0.29***	0.16***	-0.19***
	(0.05)	(0.06)	(0.05)	(0.05)
R-squared	0.499	0.801	0.413	0.703
Obs.	291	238	282	229

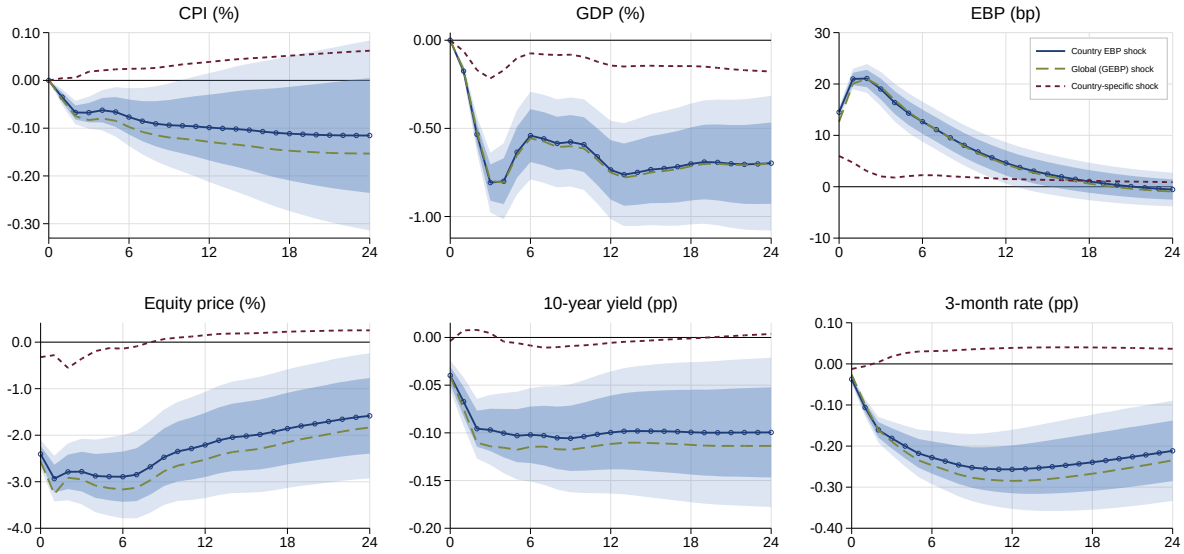
Note. The excess bond premium enters as two orthogonal terms: the global EBP, and the country-specific EBP, obtained as the residual from a regression of the country EBP on the global EBP. Forecasting regressions in the spirit of Gilchrist and Zakrajšek (2012), estimated on monthly data from January 1998 to June 2023. The dependent variable is the change in the unemployment rate (UER) or the annualized growth rate of manufacturing industrial production (IPM) over the following 3 or 12 months, as indicated in the column heading. Every regression also includes the term spread, the real policy rate, the fitted component of the corporate spread, and lags of the first difference of the outcome, whose order is selected by AIC over one to eleven lags. All variables are standardized to have mean zero and unit variance, so that the coefficients are comparable across columns. Standard errors, computed by the Newey–West estimator with four lags, are in parentheses; *, ** and *** denote significance at the 10%, 5% and 1% levels. The reported R^2 is that of the corresponding OLS regression.

Appendix G: Dynamics EBP shock

Our country-level EBP series produce intuitive results when following Gilchrist and Zakrajšek (2012) in conducting a VAR-based analysis of dynamics in various financial and macroeconomic variables following country-level EBP *shocks*. In particular, we include the country-level EBP in a VAR alongside the country's CPI, GDP, main equity index, the 10-year sovereign bond yield, and the 3-month sovereign bond yield. In line with Gilchrist and Zakrajšek (2012), we identify a shock to the EBP under the assumption that equity prices and interest rates can respond contemporaneously to EBP movements, while CPI and GDP can only respond with a one-period lag. A log-transformation is applied to the CPI, GDP, and equity prices, and these three variables enter the system in first differences: their log levels are I(1), and the level VAR is explosive in several of the countries we consider, which leaves the implied impulse responses without a well-defined long run. The system is estimated with four lags and a constant, and the responses of the differenced variables are cumulated, so that each is read as a percent deviation of the level from its no-shock path.

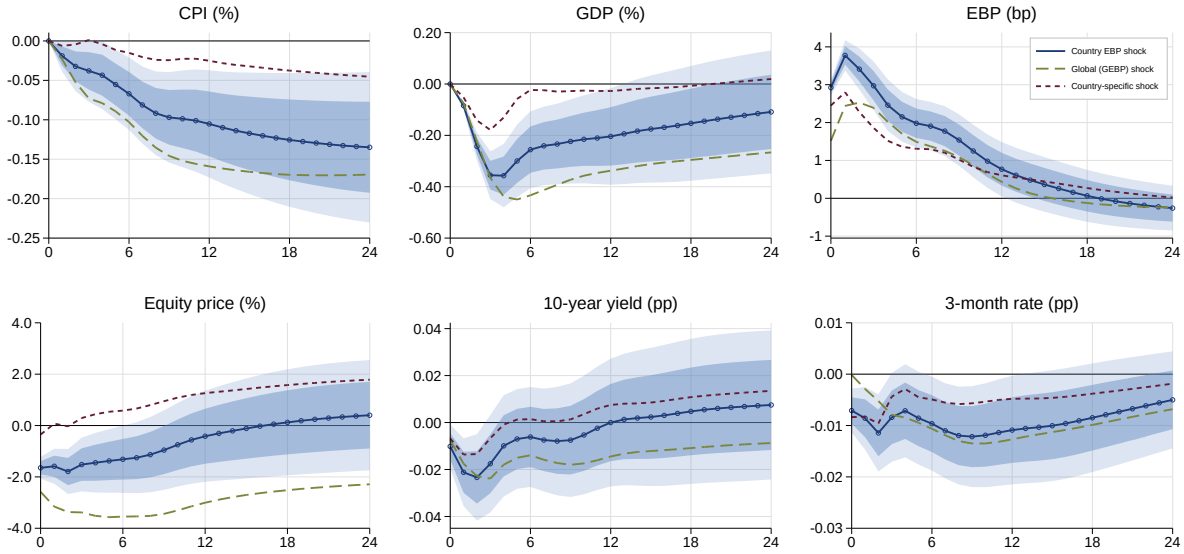
Figures G.1 to G.5 depict the resulting impulse responses to a one-standard deviation orthogonalized shock to the EBP for 5 major economies in our sample (the UK, Japan, Germany, Canada, and Australia). Across countries, we see that the EBP shock is associated with a reduction in economic output that is still present after two years. The price level falls in every country in the sample. Equity prices take an immediate hit, of between 1.6 and 2.7 percent on impact. The tightening in financial conditions and the following decline in output and prices is further accompanied by lower short-term interest rates in every country, and by lower long-term yields everywhere except Japan, where the 10-year yield is essentially unchanged throughout. These dynamics are similar across countries, and much like the original results reported for the US by Gilchrist and Zakrajšek (2012) (or the results we obtain when deploying our US-EBP series in a VAR for the US; see Figure G.6). These exercises build further confidence in our country-level EBP series appropriately capturing financial conditions across the economies in our sample.

Figure G.1 VAR Impulse Responses to a Country EBP Shock: U.K.



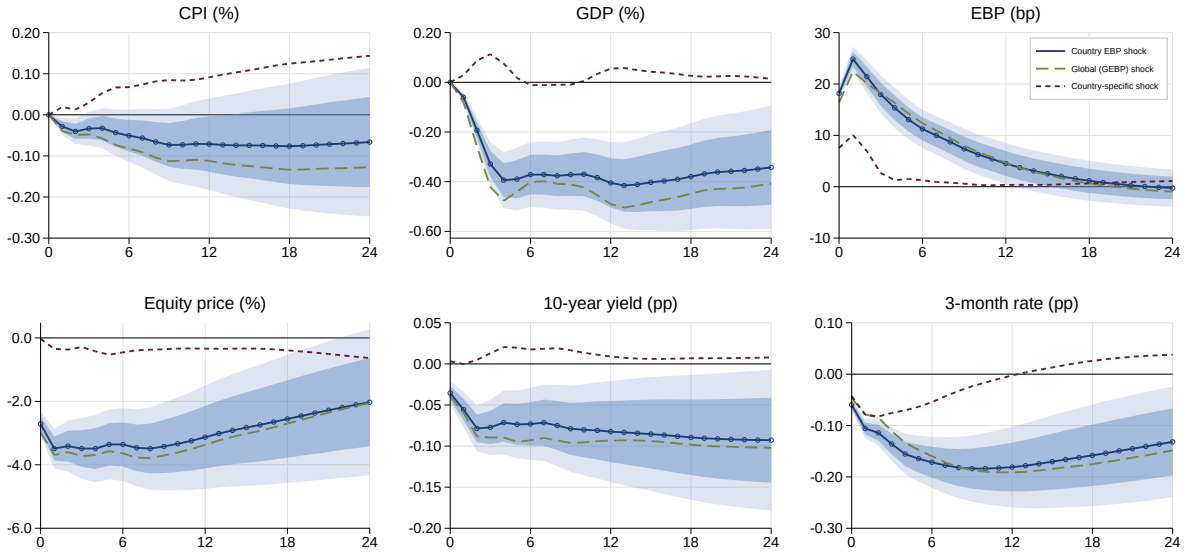
Note. Responses to a one-standard-deviation orthogonalised shock to the country EBP, in monthly steps over a 24-month horizon. The VAR contains the log first differences of the CPI, GDP and the main equity index, the country EBP, and the 10-year and 3-month sovereign yields, estimated with four lags and a constant. It is specified in first differences because the log levels are I(1), which leaves the level VAR explosive in several countries and the impulse responses without a well-defined long run. The shock is identified recursively with the EBP ordered third, so that the CPI and GDP respond only with a one-month lag while equity prices and both interest rates may respond on impact. The responses of the three differenced variables are cumulated, so that those panels read as percent deviations of the level; the EBP and the two yields are plain orthogonalised responses, in basis points and percentage points respectively. Shaded bands are 68% and 90% asymptotic (delta-method) confidence intervals. The two dashed lines split the shock into a global component, obtained by projecting the country EBP on the global EBP, and the orthogonal country-specific residual; each is a one-standard-deviation shock to its own innovation within a single re-estimated system, so the three lines are not rescaled to a common EBP impact.

Figure G.2 VAR Impulse Responses to a Country EBP Shock: Japan



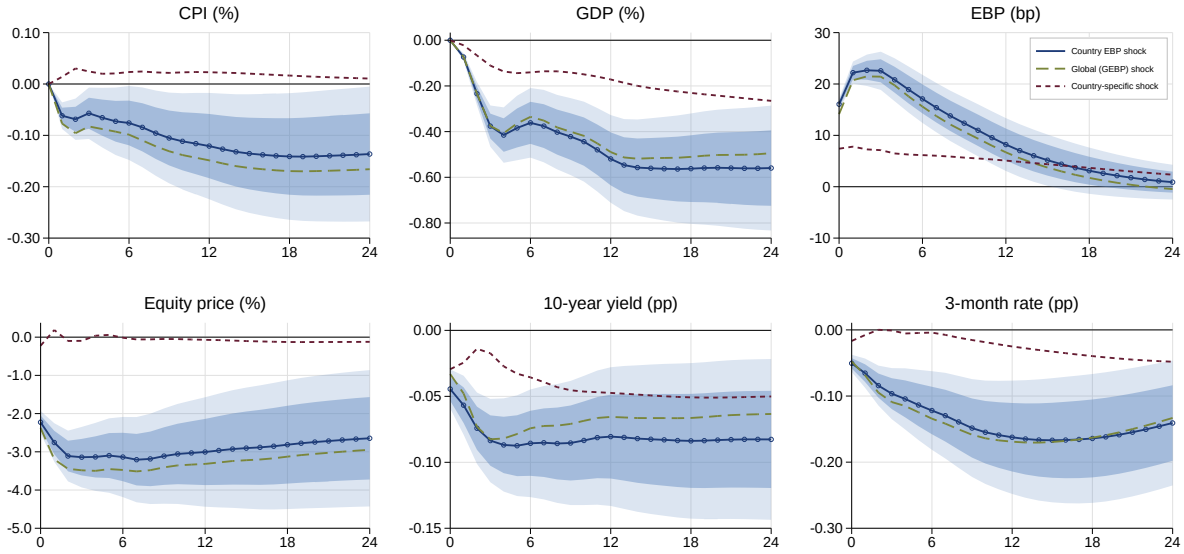
Note. Responses to a one-standard-deviation orthogonalised shock to the country EBP, in monthly steps over a 24-month horizon. The VAR contains the log first differences of the CPI, GDP and the main equity index, the country EBP, and the 10-year and 3-month sovereign yields, estimated with four lags and a constant. It is specified in first differences because the log levels are I(1), which leaves the level VAR explosive in several countries and the impulse responses without a well-defined long run. The shock is identified recursively with the EBP ordered third, so that the CPI and GDP respond only with a one-month lag while equity prices and both interest rates may respond on impact. The responses of the three differenced variables are cumulated, so that those panels read as percent deviations of the level; the EBP and the two yields are plain orthogonalised responses, in basis points and percentage points respectively. Shaded bands are 68% and 90% asymptotic (delta-method) confidence intervals. The two dashed lines split the shock into a global component, obtained by projecting the country EBP on the global EBP, and the orthogonal country-specific residual; each is a one-standard-deviation shock to its own innovation within a single re-estimated system, so the three lines are not rescaled to a common EBP impact.

Figure G.3 VAR Impulse Responses to a Country EBP Shock: Germany



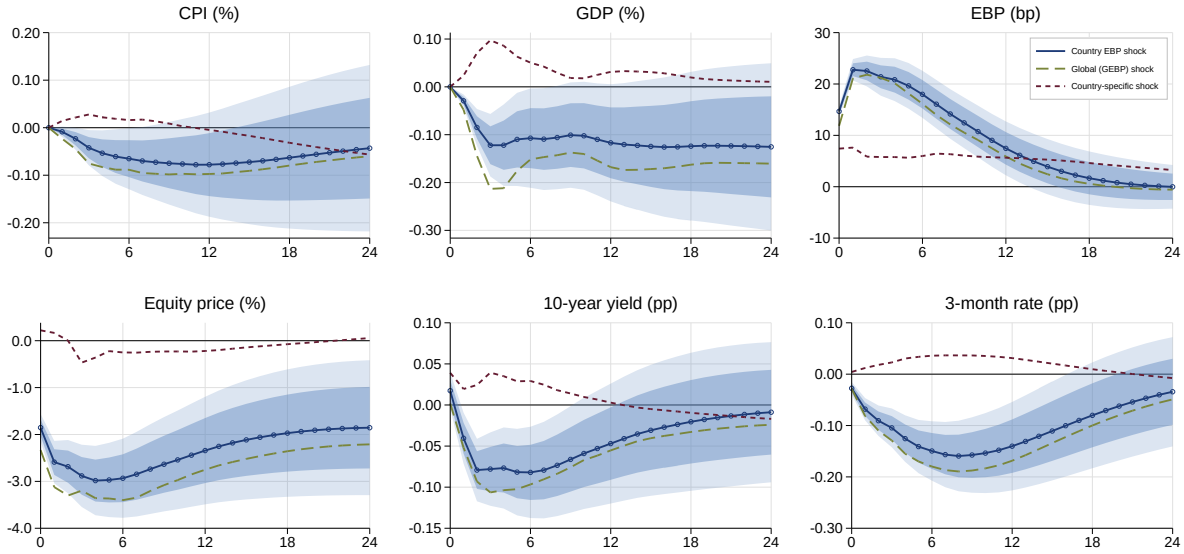
Note. Responses to a one-standard-deviation orthogonalised shock to the country EBP, in monthly steps over a 24-month horizon. The VAR contains the log first differences of the CPI, GDP and the main equity index, the country EBP, and the 10-year and 3-month sovereign yields, estimated with four lags and a constant. It is specified in first differences because the log levels are I(1), which leaves the level VAR explosive in several countries and the impulse responses without a well-defined long run. The shock is identified recursively with the EBP ordered third, so that the CPI and GDP respond only with a one-month lag while equity prices and both interest rates may respond on impact. The responses of the three differenced variables are cumulated, so that those panels read as percent deviations of the level; the EBP and the two yields are plain orthogonalised responses, in basis points and percentage points respectively. Shaded bands are 68% and 90% asymptotic (delta-method) confidence intervals. The two dashed lines split the shock into a global component, obtained by projecting the country EBP on the global EBP, and the orthogonal country-specific residual; each is a one-standard-deviation shock to its own innovation within a single re-estimated system, so the three lines are not rescaled to a common EBP impact.

Figure G.4 VAR Impulse Responses to a Country EBP Shock: Canada



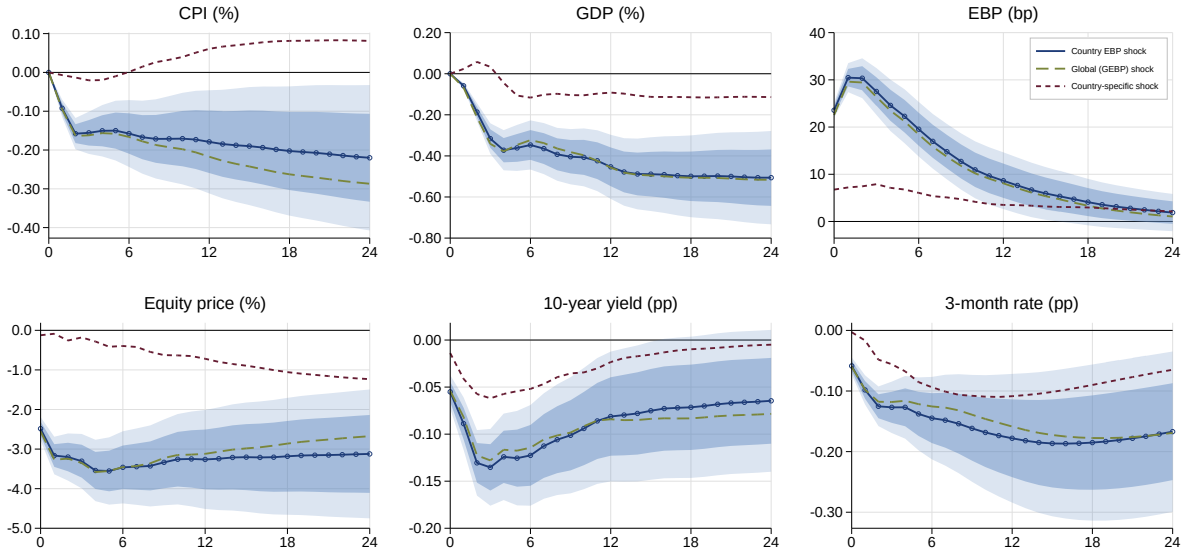
Note. Responses to a one-standard-deviation orthogonalised shock to the country EBP, in monthly steps over a 24-month horizon. The VAR contains the log first differences of the CPI, GDP and the main equity index, the country EBP, and the 10-year and 3-month sovereign yields, estimated with four lags and a constant. It is specified in first differences because the log levels are I(1), which leaves the level VAR explosive in several countries and the impulse responses without a well-defined long run. The shock is identified recursively with the EBP ordered third, so that the CPI and GDP respond only with a one-month lag while equity prices and both interest rates may respond on impact. The responses of the three differenced variables are cumulated, so that those panels read as percent deviations of the level; the EBP and the two yields are plain orthogonalised responses, in basis points and percentage points respectively. Shaded bands are 68% and 90% asymptotic (delta-method) confidence intervals. The two dashed lines split the shock into a global component, obtained by projecting the country EBP on the global EBP, and the orthogonal country-specific residual; each is a one-standard-deviation shock to its own innovation within a single re-estimated system, so the three lines are not rescaled to a common EBP impact.

Figure G.5 VAR Impulse Responses to a Country EBP Shock: Australia



Note. Responses to a one-standard-deviation orthogonalised shock to the country EBP, in monthly steps over a 24-month horizon. The VAR contains the log first differences of the CPI, GDP and the main equity index, the country EBP, and the 10-year and 3-month sovereign yields, estimated with four lags and a constant. It is specified in first differences because the log levels are I(1), which leaves the level VAR explosive in several countries and the impulse responses without a well-defined long run. The shock is identified recursively with the EBP ordered third, so that the CPI and GDP respond only with a one-month lag while equity prices and both interest rates may respond on impact. The responses of the three differenced variables are cumulated, so that those panels read as percent deviations of the level; the EBP and the two yields are plain orthogonalised responses, in basis points and percentage points respectively. Shaded bands are 68% and 90% asymptotic (delta-method) confidence intervals. The two dashed lines split the shock into a global component, obtained by projecting the country EBP on the global EBP, and the orthogonal country-specific residual; each is a one-standard-deviation shock to its own innovation within a single re-estimated system, so the three lines are not rescaled to a common EBP impact.

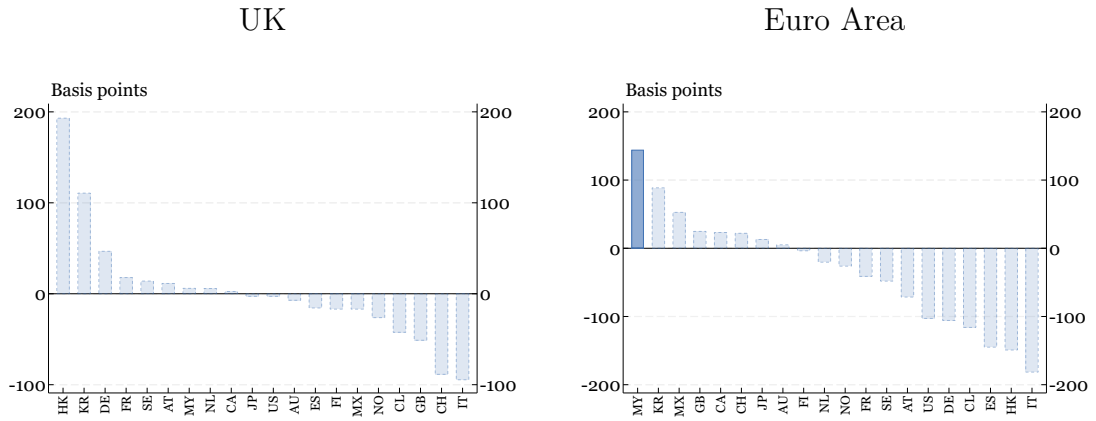
Figure G.6 VAR Impulse Responses to a Country EBP Shock: U.S.



Note. Responses to a one-standard-deviation orthogonalised shock to the country EBP, in monthly steps over a 24-month horizon. The VAR contains the log first differences of the CPI, GDP and the main equity index, the country EBP, and the 10-year and 3-month sovereign yields, estimated with four lags and a constant. It is specified in first differences because the log levels are I(1), which leaves the level VAR explosive in several countries and the impulse responses without a well-defined long run. The shock is identified recursively with the EBP ordered third, so that the CPI and GDP respond only with a one-month lag while equity prices and both interest rates may respond on impact. The responses of the three differenced variables are cumulated, so that those panels read as percent deviations of the level; the EBP and the two yields are plain orthogonalised responses, in basis points and percentage points respectively. Shaded bands are 68% and 90% asymptotic (delta-method) confidence intervals. The two dashed lines split the shock into a global component, obtained by projecting the country EBP on the global EBP, and the orthogonal country-specific residual; each is a one-standard-deviation shock to its own innovation within a single re-estimated system, so the three lines are not rescaled to a common EBP impact.

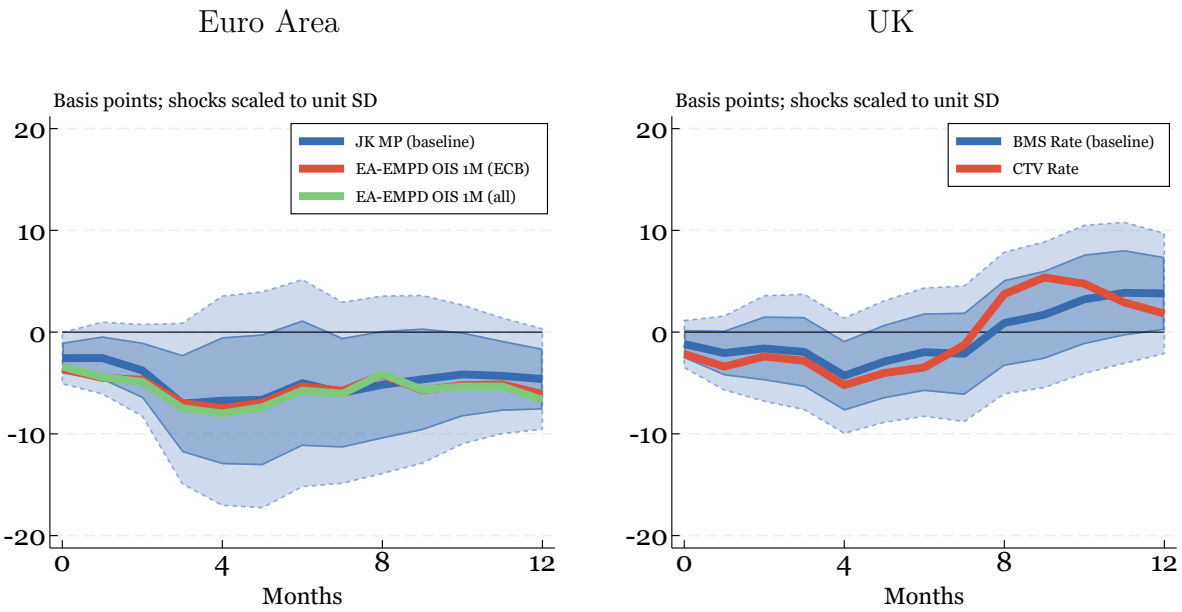
Appendix H: Additional monetary policy results

Figure H.1 Country-Level Responses of the EBP to UK and Euro Area Monetary Policy Shocks



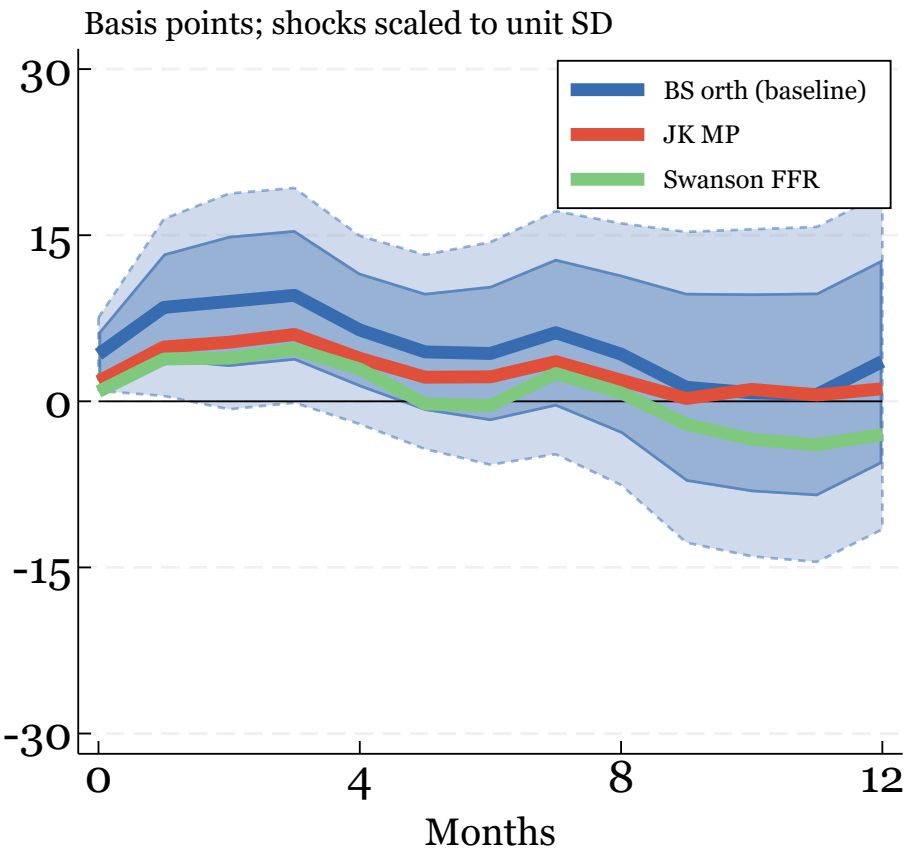
Notes: Country-specific responses to a UK and EA monetary policy shock calculated in a country-level regression for a one-month horizon. Filled bars indicate significance of the coefficient on a 10% significance level using heteroskedasticity-robust standard errors.

Figure H.2 Response of Euro Area and U.K. EBP to alternative local Monetary Policy Shocks



Notes: Dynamic response of the EA EBP to various euro area shocks and of the U.K. EBP to various U.K. monetary policy shocks. For the euro area the red and green lines correspond to alternative shocks taken from the Euro Area Extended Monetary Policy Event-Study Database of Altavilla et al. (2026). The OIS-1m response is taken as rate surprise. The red line only includes events that are part of the original Euro Area Monetary Policy Event-Study Database while the green line corresponds to shocks from all events, including speeches. For the UK, the red line corresponds to monetary policy shocks of Cesa-Bianchi et al. (2020), which are available until 2015 only. Shaded areas correspond to the 68% and 90% confidence intervals for the baseline shocks, using heteroskedasticity-robust standard errors.

Figure H.3 Response of the GEBP to various US Monetary Policy shocks



Notes: Dynamic response of the GEBP worldwide to various US monetary policy shocks. Alternative shocks correspond to Jarociński and Karadi (2020) (red) and Swanson (2021) (green). Shaded areas correspond to the 68% and 90% confidence intervals of the baseline shock, where heteroskedasticity-robust standard errors are used.

Table H.1 Bond-level EBP response to various monetary policy shocks by currency

Currency (share of bond-months)	US shock	ECB shock	BoE shock
USD (74%)	10.8** (5.2)	2.6 (3.6)	−3.9 (5.9)
EUR (16%)	8.3** (3.5)	−0.2 (2.3)	−0.8 (3.4)
GBP (5%)	6.2* (3.4)	2.1 (2.0)	−1.0 (4.8)
JPY, CAD and AUD (6%)	1.9* (1.1)	1.5 (1.2)	0.3 (1.7)

Note. Currency-specific coefficient α on shock j as estimated in

$$y_{b,t+h} = \alpha_b + \alpha^j_{\text{curr}(b)}\varepsilon^j_t + \sum_{l=1}^{12}\rho_l y_{b,t-l} + e_{b,t+h}.$$

for $h = 1$. Standard errors are clustered at the firm and month level. *, ** and *** denote significance at the 10%, 5% and 1% levels.

Table H.2 Bond-level EBP response to various monetary policy shocks by currency and issuer country

Sample	US shock	ECB shock	BoE shock
All bonds (598k)	9.6** (4.5)	2.1 (3.1)	−2.9 (5.0)
EUR bonds, EA issuers (68k)	8.2** (3.8)	−1.0 (2.3)	−0.1 (3.1)
GBP bonds, UK issuers (16k)	6.1* (3.6)	1.4 (1.6)	−0.9 (3.8)
USD bonds, US issuers (347k)	10.1** (4.9)	2.2 (3.2)	−4.7 (6.2)
JPY bonds, JP issuers (12k)	0.4** (0.2)	0.1 (0.4)	−0.3 (0.3)

Note. Sample-specific coefficient α on shock j as estimated in

$$y_{b,t+h} = \alpha_b + \beta^j\varepsilon^j_t + \sum_{l=1}^{12}\rho_l y_{b,t-l} + e_{b,t+h}.$$

for $h = 1$. Standard errors are clustered at the firm and month level. *, ** and *** denote significance at the 10%, 5% and 1% levels.